

Always-On Personal AI: A Roadmap towards a Continuing and Self-Evolving Loop

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Always-on personal AI sustains assistance between explicit requests under a declared schedule and availability contract, retaining usable user-specific state across interactions. This *continuing loop* chooses silence, communication, preparation, or execution and carries unresolved commitments across interaction boundaries. Governed *self-evolution* extends this baseline by using attributed outcomes to propose, validate, and durably commit policy revisions. We present a functional architecture and roadmap organized around *Sensing*, *Memory*, and *Action*, five coupling channels, and shared conditions of privacy, resources, uncertainty, and user authority. We compare each module’s spaces and policies, examine their dependencies in integrated systems, and distinguish cumulative levels of loop closure. The literature reveals an *integration gap*: component capabilities and selected couplings have advanced, while evidence of reliable joint operation and lasting benefit remains limited. Two priorities follow: preserving opportunities to detect change under memory-guided observation, and establishing when outcome feedback justifies persistent updates. To support reproducible assessment, we introduce the *Harness for Always-on Loop Oversight* (HALO), with read-only review and scoped execution modes. This paper’s read-mode review of 31 configurations links channel mechanisms to source locations and separates implementation status from supporting evidence. The resulting research agenda connects corrective evidence, governed updates, and later user benefits and costs.

<https://github.com/choiszt/awesome-always-on-personal-ai>

<https://Always-On-Agent.github.io>

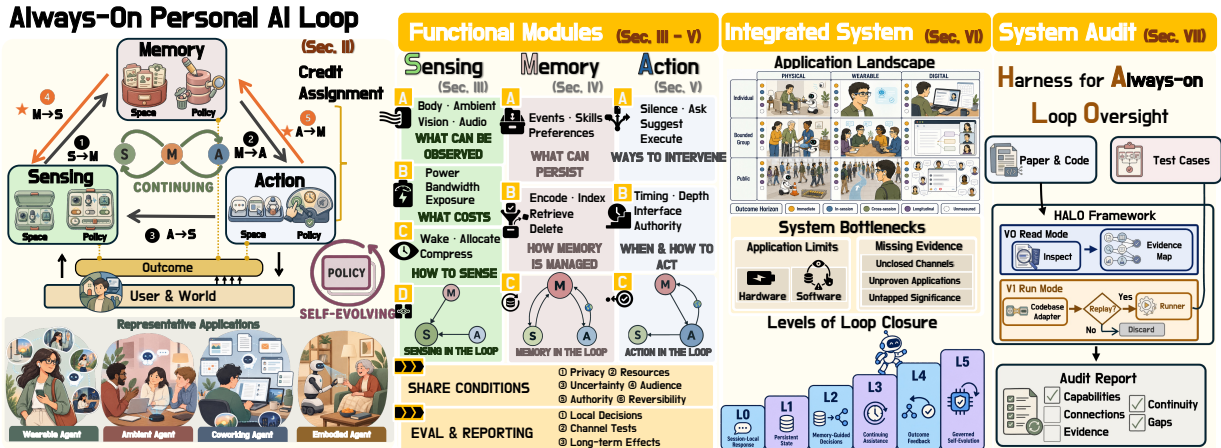


Figure 1: **Overview of the roadmap.** We introduce a policy-coupled Sensing–Memory–Action framework and describe its runtime information flow (Sec. 2). Highlighted $M \rightarrow S$ and $A \rightarrow M$ arrows indicate higher relative difficulty under changing user needs, limited resources, and incomplete feedback. Building on this framework, we review each module’s space, policies, and coupling channels (Secs. 3–5). We then examine applications, integration bottlenecks, and loop closure from a system-level perspective (Sec. 6). HALO supports read (V0) and run (V1) modes (Sec. 7); the system comparisons in this paper come from read mode.

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1 Introduction

The most profound technologies are those that disappear. They weave themselves into the fabric of everyday life until they are indistinguishable from it.

Weiser [1]

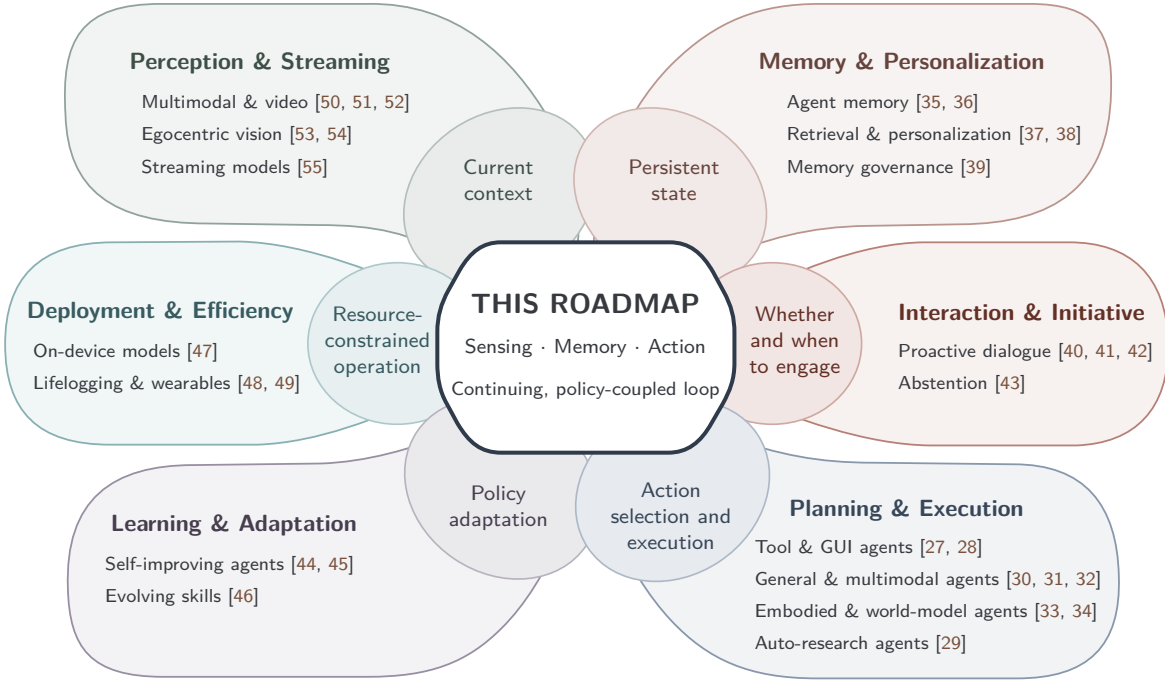
Personal AI is moving beyond isolated requests toward assistance that uses the user’s current situation, goals, and past interactions to decide whether and how to help. Research toward this form of assistance spans physical, wearable, and digital settings. In physical settings, embodied assistants [2, 3, 4, 5, 6] carry out tasks and coordinate assistance with human partners. Wearable assistants [7, 8, 9, 10] access live egocentric context during everyday activities. Digital assistants [11, 12, 13, 14, 15] use computer activity and application events to anticipate user needs. Across these settings, assistance needs to build on earlier interactions rather than start afresh with each request [16, 17, 18].

This raises a question: what must continue when an interaction ends? Task-driven agents [19] are commonly evaluated in bounded episodes across web and desktop environments [20, 21] and embodied household tasks [22, 23]. Success within an episode does not establish service across interactions. An always-on personal assistant must both carry relevant personal state forward and remain able to monitor or resume assistance between explicit requests. A conversation may end while a task is still pending or an earlier action’s outcome is unresolved. We call this operating baseline a *continuing loop*: decisions recur across interactions under a declared schedule and availability contract, and an interaction boundary does not itself reset the state, commitments, or unresolved outcomes that later decisions depend on.

Beyond this continuing baseline, a long-term goal is a *self-evolving loop* that revises its decision policies using attributed outcomes. A continuing loop can preserve and update personal state while these policies remain fixed. A reminder may be dismissed because it interrupts a meeting, because its content is unhelpful, or because the task has already been completed elsewhere. The dismissal alone does not determine whether to correct a task record, update a preference, or revise the reminder policy. Attribution guides the choice of update; validation determines whether a proposed policy revision should be durably committed.

Realizing these goals requires more than strong individual capabilities. Multi-session evaluations [24, 25] already connect memory to later action, while memory-policy learning [26] studies feedback and credit assignment. Yet evidence of reliable, useful service over time remains limited, and studies differ in their available context, retained state, tasks, and feedback. This *integration gap* involves specific dependencies. Retained state supports continuity but can carry outdated assumptions into later decisions. When memory guides sensing, those assumptions may suppress observations needed to correct them. When outcomes guide persistent updates, delayed or ambiguous feedback can reinforce an ineffective decision rule. Understanding these failures requires tracing which information shaped a decision, what evidence could correct it, and which updates that evidence justifies.

We present a functional architecture and roadmap that organizes this analysis around *Sensing*, *Memory*, and *Action*, five coupling channels, and shared conditions involving privacy, resources, uncertainty, and user authority. The framework separates available capabilities from the policies that select and use them, then traces dependencies across these decisions. Our literature synthesis finds progress in memory-conditioned behavior and selected feedback mechanisms, but limited evidence for correcting stale observation priors and using attributed outcomes to justify durable revisions (Sections 6 and 7). These gaps motivate a roadmap from component capabilities, through the dependencies a continuing loop requires, to the evidence a self-evolving loop must additionally supply. To support reproducible assessment, we introduce the *Harness for Always-on Loop Oversight* (HALO), a common protocol with read-only review and scoped execution modes. In read mode, HALO links channel mechanisms to source passages and assesses reported traces, comparisons, and outcomes using shared evidence criteria. This produces a review record of what each study supports and which questions require further tests.



Shared research topics, not equivalent system scope or evidence.

Figure 2: **Topical overlap with capability-focused surveys.** Outer groups show representative survey themes, while inner labels identify topics shared with this roadmap. Overlap indicates shared topics rather than equivalent system scope or evidence. Concurrent system-level surveys are compared in Table 1.

1.1 Scope and Positioning

This roadmap examines personal-AI *systems and applications* across physical, wearable, and digital settings, using a common functional framework to compare their decisions and dependencies. Personal service centers on a particular user’s state, goals, and preferences. We include component and task-bounded studies when they provide mechanisms or evidence relevant to continuing personal service; inclusion does not imply that the evaluated system satisfies the full always-on definition. Detailed coverage of low-level mechanisms, such as visual recognition and robot control, is left to specialized surveys.

The intended readers are researchers designing or evaluating personal assistants and their supporting mechanisms. This is a narrative, mechanism-oriented synthesis: inclusion follows relevance to the declared functions, dependencies, and service conditions, rather than an exhaustive publication count. Foundational work supplies the conceptual lineage; contemporary papers and product documentation supply mechanisms with their reported evaluation scope. The cited snapshot includes material available by 10 September 2026. Product descriptions, simulations, and controlled studies retain distinct evidence status; targeted searches do not justify claims of field-wide absence.

We position related surveys by their primary unit of analysis. Capability-focused surveys organize literature around particular capabilities and mechanisms, while concurrent system-level surveys assemble several of those mechanisms around persistent state, device platforms, human trajectories, or model–harness co-evolution.

1.1.1 Capability-Focused Surveys

Figure 2 summarizes overlap with capability-focused surveys. These provide detailed accounts of perception across modalities and deployment settings [50, 51, 53, 54, 49, 55, 48, 47], retained information and its use and governance [35, 39, 37, 38], and proactive assistance [40, 41, 42] and agent decision and execution

mechanisms [30, 31, 32, 27, 28, 33, 34]. Self-evolving and dynamic-skill surveys further examine changes to models and agent control mechanisms [44, 56, 45, 46]. Several also discuss integration. We use these analyses to examine decisions and dependencies in continuing personal service.

Table 1: **Concurrent surveys: scope and organizing questions.** Icons indicate deployment coverage:  physical,  wearable, and  digital. Research focus identifies what each survey studies; organizing framework describes how it organizes and analyzes the literature.

Survey	Domain	Research focus	Organizing framework
<i>Always-On Agents</i> [57]	  	Persistent-state agents	State types, lifecycle, and governance axes
<i>AI Smart Glasses</i> [58]		Wearable intelligence	Hardware, intelligence, interaction, and applications
<i>ComBodied Agents</i> [59]	  	Longitudinal human support	Human-state trajectories, personal world models, and interventions
<i>From Seeing to Acting</i> [60]		First-person intelligence	Capability levels and deployment requirements
<i>Long-Horizon Agents</i> [61]	 	Long-horizon agent capabilities	Model-harness co-evolution across task horizons
This roadmap	  	Always-on personal systems	Policy-coupled Sensing–Memory–Action loops

1.1.2 Concurrent System-Level Surveys

Table 1 compares this roadmap with concurrent system-level surveys. These surveys cover overlapping topics but organize the literature around different questions.

Our view complements these perspectives by tracing dependencies between observation, retention, and action decisions. For a personal memory, we examine not just how it is stored, updated, and removed, but whether it changes subsequent observation, influences an action, and receives outcome evidence that supports further revision. This makes implementations with different sensors, memory representations, and controllers comparable at the dependency they support, while keeping service goals, operating conditions, and evaluation scope explicit. It also reveals when another mechanism can compensate for a missing or faulty dependency. Ding et al. [57] define always-on agents in terms of durable state that influences later behavior, a scope that includes agents invoked only when a user opens a session. We require both durable personal state and the ability to monitor or resume service between explicit requests.

1.2 Contributions

This roadmap makes three contributions:

- **A functional architecture for always-on personal AI.** We organize Sensing, Memory, and Action through their spaces, local policies, five coupling channels, and shared conditions. Separating capabilities from their selection and use enables comparison of dependencies across implementations.
- **A system-level synthesis and roadmap.** We explain why dependencies pose different challenges and under which conditions their failure degrades service or alternative mechanisms compensate. Memory-guided observation and outcome-driven revision illustrate the resulting bottlenecks and evidence gaps, informing priorities for continuing and self-evolving loops.
- **HALO for reproducible oversight.** We provide read and run modes under a common oversight protocol. The read-mode results presented here link all five channel statuses to source locations and distinguish mechanism descriptions, traced use, controlled effects, and controlled service outcomes. A versioned ledger makes comparisons inspectable and supports consistent review of further studies.

2 Always-On Personal AI as a Policy-Coupled Loop

In this section, we define the service studied in the roadmap and specify its functional modules, available choices, and decision dependencies. We then model continuing operation and governed self-evolution and use these concepts to locate applications and prior work.

Definition: Always-on personal AI

We define always-on personal AI as a personal service with two core properties: *operational continuity* and *state durability*. Operational continuity enables the system to monitor context, resume tasks, and decide whether to act between explicit requests under a declared schedule and availability contract. State durability preserves user-specific state for later use and carries unresolved commitments across interaction boundaries. A self-evolving loop extends this continuing baseline by using attributed outcomes to guide policy revisions and durably retaining changes that pass validation.

An invocation-only assistant with persistent personal state provides durability without establishing between-request service. A continuously running assistant that clears personal state at each session provides operational continuity without state durability. An event-triggered assistant that preserves personal state and resumes the declared service can satisfy both with fixed policies. These distinctions specify the service boundary independently of policy learning.

2.1 The Three Functional Modules

The decomposition builds on accounts of anticipation through internal models [62], the explicit roles of perception, memory, and action selection in Soar [63], and controlled information acquisition in active perception [64]. Modern precedents include LeCun’s autonomous machine intelligence proposal [65], with perception, short-term memory, an actor, and predictive and control components, and CoALA [66], with memory, internal and external actions, and decision-making.

Building on this work, we distinguish three functional modules: *Sensing*, *Memory*, and *Action*. These are functional roles: a controller may perform several, and a role may span components.

Sensing Sensing gathers and interprets physical and digital observations to construct the current state available for decision-making. Audiovisual streams, sensor readings, and application events provide context about the user and environment, accompanied by observation time, provenance, and uncertainty. Continuous or event-triggered observation keeps this context current within privacy and resource budgets. Temporary buffers support this process; information retained for use across interactions serves as Memory.

Memory Memory maintains user-specific state across interactions, including facts, preferences, past events, and unresolved commitments. New evidence and changes in the user’s circumstances provide candidate *deltas* for persistent state, and reads make retained information available to later decisions. This role depends on persistence and subsequent use, rather than whether an implementation names a database “memory.” Stored information remains open to correction and removal under privacy and retention conditions.

Action Action determines whether, when, and how the assistant helps. It uses current context, retained information, and user goals to choose silence, communication, preparation, or execution through digital tools or physical devices. Decisions may arise from an explicit request or ongoing context; choosing silence is part of the action space. Planning and predictive models support multi-step activities and assessment of consequences [34, 67]. Each intervention respects current authority and resource limits.

2.2 Spaces, Local Policies, and Shared Conditions

Throughout this roadmap, *space* describes the observations available to Sensing, the information Memory can retain, and the behavioral options available to Action. A *local policy* selects and uses these options within a module, including sampling and compression, encoding and retrieval, or behavior selection. Consider an

assistant with a camera and calendar access. Available images, calendar events, resolutions, and sampling rates form part of its sensing space. The local sensing policy decides whether to sample and at which resolution. Shared conditions govern whether collection and retention are permitted and how much resource the decision may consume.

Shared conditions. Shared conditions apply across the loop, shaping both local decisions and exchanges between modules. These include privacy requirements, resource budgets, uncertainty, and user authority. For example, during a confidential meeting, a privacy rule may allow real-time speech processing but prohibit saving a transcript or sending meeting content to external tools. The same rule permits speech processing in Sensing, restricts persistent storage in Memory, and limits external disclosure through Action.

Muse [15] illustrates how user authority constrains multiple modules. Its permission-checking mechanism, Sentinel, checks requests to connected services and external websites against the user’s permissions, allowing, denying, or seeking approval as appropriate [68]. A user may allow calendar access to detect conflicts while withholding permission to schedule meetings [68]. This boundary constrains both the context Sensing may obtain and the changes Action may make. Remembered preferences or permission records can inform such decisions, but cannot create authority beyond the current authorization boundary.

2.3 Five Coupling Channels

Local decisions can depend on information or decisions from other modules. We describe these directed dependencies as *coupling channels*. A *coupling policy* specifies the information or control signal supplied across a functional boundary; the receiving local policy uses it with current inputs and shared conditions to make its decision. For example, a coupling policy can select and interpret remembered routines to guide Sensing, while the local sensing policy decides whether and when to observe. This distinction identifies information sources, decision responsibilities, and intervention points. An implementation may combine these decisions in a single computation. Among Sensing, Memory, and Action, six directed cross-module pairs can be distinguished. We treat $S \rightarrow A$ as the basic sense-and-respond path and focus on the remaining five. The baseline and its costs remain part of the local Sensing and Action analyses; the service determines which further dependencies are needed.

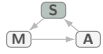
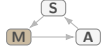
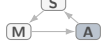
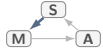
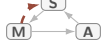
Definitions of the five channels

- $S \rightarrow M$ selects sensed evidence for persistent admission. Its coupling policy requests the write; the local Memory policy realizes the state change. Admission may be user-directed or autonomous.
- $M \rightarrow A$ uses retained personal state to guide whether, when, and how the system acts within current grants of authority. For example, a preference for quiet mornings can defer routine reminders.
- $A \rightarrow S$ uses action intentions, plans, or execution state to guide subsequent observation, including information needed before acting and checks of expected effects [69]. A planned grasp can direct attention to a cup’s handle; execution can prompt a check that the cup was lifted.
- $M \rightarrow S$ uses retained state to guide observation and interpretation. Remembered routines can shape sampling and attention; personal priors can change how the same observations are understood.
- $A \rightarrow M$ links externally observed consequences, including those of silence, to earlier decisions and uses that attribution to govern persistent changes to personal state or decision policies.

In the reminder example, an assistant retains a pending task ($S \rightarrow M$) and uses a preference for quiet mornings to defer a reminder ($M \rightarrow A$). A planned reminder prompts a check that a call has ended ($A \rightarrow S$), while remembered routines guide when the assistant checks availability ($M \rightarrow S$). After a dismissal, $A \rightarrow M$ links the feedback to the earlier decision and identifies which persistent state or policy that evidence justifies revising. This use of feedback connects the channels to closed-loop control [70].

A functional dependency can also be realized through several existing channels working together. For example, remembered preferences can help Action select a question ($M \rightarrow A$), and that question determines what

Table 2: **The framework: three modules and five coupling channels.** Local policies govern choices within modules; coupling policies govern dependencies between them. Core challenges summarize their sources of difficulty. Miniature loops locate modules and channels; coral dashed edges mark High difficulty.

A. Functional modules			
Module	Space	Local policies	Core challenge
Sensing 	Observations, events, current-state representations	π_S : sample; allocate attention; filter; segment; compress; interpret	Construct useful current context from partial, noisy streams within resource and privacy budgets.
Memory 	Persistent events, facts, preferences, tasks, skills, summaries, provenance	π_M : encode; index; retrieve; consolidate; correct; retire	Maintain relevant, consistent, and correctable personal state as facts and preferences change.
Action 	Silence, communication, preparation, execution	π_A : gate intervention; select timing; select control depth and effector within granted authority	Choose useful, timely effects while respecting authority, uncertainty, and reversibility.
B. Coupling channels			
Channel	Coupling policies	Difficulty*	Core challenge
$S \rightarrow M$ 	κ_{SM} : admit sensed evidence and request creation or refresh; π_M realizes the write	Moderate	Future usefulness is uncertain at admission, while missed evidence may be irrecoverable and excess retention costly.
$M \rightarrow A$ 	κ_{MA} : use personal state to condition action timing, content, effector, or commitment	Moderate	Personal records may be stale or conflicting; their use must fit current context, timing, and authority.
$A \rightarrow S$ 	κ_{AS} : turn action intentions, plans, and execution state into observation targets, including checks of expected effects	Moderate	Action-relevant uncertainty must be translated into timely observations despite partial observability and changing plans or execution state.
$M \rightarrow S$ 	κ_{MS} : use retained priors to guide sampling, attention, compression, or interpretation	High	A prior shapes which evidence is acquired; stale priors can suppress observations needed to correct them.
$A \rightarrow M$ 	κ_{AM} : attribute outcomes to decisions; govern memory or policy updates	High	Sparse, delayed outcomes mix several causes; attribution must justify lasting updates.

* Ratings reflect our qualitative assessment of relative difficulty under changing user needs, resource constraints, and incomplete feedback. Sections 3–5 develop the underlying challenges.

Sensing needs to observe ($A \rightarrow S$), allowing memory to guide observation through the action decision. Channels may also share evidence: the same dismissal record can first be admitted as observed evidence through $S \rightarrow M$ and later support attribution through $A \rightarrow M$.

The difficulty differs across these dependencies. In $M \rightarrow S$, a retained prior can change which evidence becomes available, so an outdated routine that suppresses observation may also suppress its own correction. In $A \rightarrow M$, outcomes can arrive after other actions or changes in the user’s circumstances; attributing a dismissal to timing must distinguish these alternatives before justifying a lasting change. Evidence sufficient to correct one task record may still be insufficient to revise a general reminder policy. Table 2 assigns High to these two channels and Moderate to the other three under changing user needs, limited budgets, and incomplete feedback. These are qualitative judgments about those conditions, rather than fixed difficulty levels across all services or maturity rankings based on publication counts.

Their effect on service is also conditional. Under a tight sensing budget, stale $M \rightarrow S$ priors may cause missed opportunities for correction; a service with sufficiently broad observation may tolerate weaker memory guidance. A reliable tool receipt may satisfy a verification need without an actively targeted $A \rightarrow S$ check. Analysis therefore identifies the evidence the service needs and whether the available mechanisms can provide

it. When information is not passed on or an update is not made, the relevant mechanism may be absent, an existing mechanism may have failed, or privacy and permission rules may have blocked the operation. For example, withholding sensitive information when storage is prohibited or deferring an update when feedback is ambiguous can be appropriate decisions. If the paper provides insufficient implementation details or evidence of an effect, the corresponding claim remains unverified.

2.4 Continuing Operation and Governed Self-Evolution

The interfaces below locate where information enters a decision and where a policy revision becomes durable. The module sections follow the receiving decision: Sensing examines $M \rightarrow S$ and $A \rightarrow S$ (Section 3); Memory examines $S \rightarrow M$ and $A \rightarrow M$ (Section 4); Action examines $M \rightarrow A$ (Section 5). This organization connects each dependency to its local policy, failure modes, and evaluation.

Continuing operation. Let t index decision opportunities, including those generated by the declared service schedule or an external event without a new user request. The system carries forward personal memory m_t , an observation target y_t^{AS} from earlier action decisions, the policy implementation θ_t , and the available execution trace $\tau_{<t}$. They form the persistent state $z_t = (m_t, y_t^{AS}, \theta_t, \tau_{<t})$. Here m_t supplies information about the user and tasks, while θ_t implements the rules governing observation, memory management, and action. We denote these policies by Π_{θ_t} : the three local policies $\pi_S^{\theta_t}$, $\pi_M^{\theta_t}$, and $\pi_A^{\theta_t}$, and the five coupling policies $\kappa_E^{\theta_t}$.

At decision t , inputs $\xi_t = (x_{\leq t}, u_t, o_t)$ comprise accessible observation history, an optional explicit request, and optional directly delivered feedback, such as a user response or tool receipt. A request requiring interpretation or retention is also represented in the accessible input stream $x_{\leq t}$, so it can reach s_t and admission; u_t marks its explicit role in Action. The notation $x_{\leq t}$ does not grant permanent, cost-free access to all raw observations: cross-interaction access requires a declared retention and access mechanism, governed as persistent state. Similarly, $\tau_{<t}$ may contain bounded or summarized traces under retention rules. Under shared conditions c_t , the system transition $\mathcal{F}$ constructs a current sensed state s_t , selects an action a_t , and updates persistent state:

$$\begin{aligned} z_{t+1} &= \mathcal{F}_{\Pi_{\theta_t}}(z_t, \xi_t; c_t), & t \in \mathbb{N}, & \quad \mathcal{Z}_{\text{term}} = \emptyset, \\ b_t = 1 &\nRightarrow (m_{t+1}, \tau_{\leq t}) = (m_{\text{init}}, \emptyset). \end{aligned} \tag{1}$$

The first line models active service without an automatic task-terminal reset; user termination, deletion, and governed reset remain possible. The second states that an application-defined session boundary b_t does not itself clear personal state and traces. Duty cycling, event-triggered wake-up, planned pauses, and governed recovery can preserve operational continuity and state durability, with explicit handling of unresolved commitments and external effects. Allowing $u_t = \emptyset$ represents request-free decision opportunities, but a concrete system must demonstrate the scheduling or trigger mechanism that creates them. At any opportunity, the action policy may choose silence.

Reinforcement learning makes the same distinction between episodic tasks, which terminate and reset, and continuing tasks, which do not [71]. Long-run average net utility provides an evaluation objective for such a process. This separates always-on operation from an extended multi-turn episode and explains *episodic amnesia* [72]. Reset-free reinforcement learning [73], average-reward mobile-health intervention [74, 75], and long-horizon recommendation under sparse delayed signals [76, 77] instantiate the same modeling choice. No finite evaluation window certifies long-run performance: a write useful today may become stale, while a successful intervention may spend attention that matters more later.

Components of the continuing loop. The local and coupling policies in Π_{θ_t} realize each system transition $\mathcal{F}$. They can be implemented through learned models, symbolic rules, retrieval operators, or tool orchestration. Shared conditions, including privacy, resources, uncertainty, and user authority, may restrict a local operation or leave a channel inactive at a decision. The three local policies construct the current sensed state, update personal memory, and select an action:

$$\begin{aligned} s_t &= \pi_S^{\theta_t}(x_{\leq t}, y_t^{MS}, y_t^{AS}; c_t), \\ m_{t+1} &= \pi_M^{\theta_t}(m_t, y_t^{SM}, y_t^{AM}; c_t), \\ a_t &= \pi_A^{\theta_t}(s_t, y_t^{MA}, u_t, \hat{o}_t; c_t), \end{aligned} \quad (2)$$

where $y_0^{AS} = \emptyset$, $\hat{o}_t$ denotes anticipated consequences of candidate actions, and y_t^E denotes a coupling signal. These signals also identify where a controlled comparison can vary an input to a receiving decision. The Memory equation describes the state-update part of π_M ; read operations and candidate retrieval are also local Memory decisions, whose results supply the outgoing coupling policies. We write those policies as functions of m_t without expanding the read interface. The signals are an admitted write request y_t^{SM} , action-conditioning information y_t^{MA} , an observation target y_t^{AS} , a sensing prior y_t^{MS} , and attributed feedback y_t^{AM} :

$$\begin{aligned} y_t^{SM} &= \kappa_{SM}^{\theta_t}(s_t, m_t; c_t), & y_t^{MA} &= \kappa_{MA}^{\theta_t}(m_t, s_t; c_t), \\ y_{t+1}^{AS} &= \kappa_{AS}^{\theta_t}(a_t, \hat{o}_t; c_t), & y_t^{MS} &= \kappa_{MS}^{\theta_t}(m_t, x_{\leq t}; c_t), \\ y_t^{AM} &= \kappa_{AM}^{\theta_t}(s_t, o_t, a_{<t}, \tau_{<t}; c_t). \end{aligned} \quad (3)$$

These interfaces separate admission from mutation: κ_{SM} requests a write and π_M realizes any persistent state change. Outgoing reads and action use m_t , while current writes produce m_{t+1} . This is one causal ordering; fused, asynchronous, or read-after-write implementations can be described using finer decision opportunities. Preparation, clarification, and final execution may therefore occupy separate opportunities, with newly acquired evidence and current permissions checked before commitment.

At decision t , consequences of earlier actions enter through sensed state s_t or directly delivered feedback o_t . Environmental changes follow $x_{\leq t} \rightarrow \pi_S \rightarrow s_t$; user responses or tool receipts may arrive directly. The policy κ_{AM} uses this evidence and $\tau_{<t}$ to link outcomes to earlier actions $a_{<t}$. Internal self-assessment can suggest explanations, whose support must come from verifiable outcome evidence. The available evidence may justify a personal-state update through π_M , a policy-revision proposal, both, or neither.

Memory can retain past experience and future-facing plans [78], and Action considers anticipated consequences $\hat{o}_t$ [79]. Actual feedback, however, may arrive many decision opportunities later; retained traces can preserve the intervening context needed to interpret it. Channels can operate selectively as evidence arrives.

Self-evolving extension: governed policy revision. The continuing loop can preserve and update personal state while keeping its local and coupling policies fixed. The self-evolving loop adds governed self-evolution: the use of attributed outcomes to propose, validate, and durably commit revisions to these policies. Revisions may change model parameters or control code and configurations outside the models:

$$\begin{aligned} \varepsilon_t &= \mathcal{E}(\Pi_{\theta_t}, y_t^{AM}, \tau_{\leq t}; c_t), \\ \Delta\theta_t &= \mathcal{U}_{\psi_t}(\theta_{1:t}, \varepsilon_t; c_t), \\ \theta_{t+1} &= \begin{cases} \theta_t, & \text{fixed-policy continuing loop,} \\ \text{Commit}(\theta_t, \Delta\theta_t \mid V_{\phi_t}, c_t), & \text{self-evolving extension.} \end{cases} \end{aligned} \quad (4)$$

Here $\mathcal{E}$ turns execution and attributed feedback into update evidence, $\mathcal{U}_{\psi_t}$ proposes a revision from the policy history $\theta_{1:t}$ and this evidence, and the verifier V_{ϕ_t} and conditions c_t govern whether it is durably committed; rejection returns θ_t . Statistical acceptance tests can limit the risk of committing an ineffective revision [80, 81]. Its benefit is measured by whether the agent behaves better when using it [82]. This factorization echoes recent self-improvement work and makes the persistence gate explicit.

To specify what can change, we write $\theta_t = (\omega_t, \Sigma_t)$, following the model-scaffold distinction in recent agent self-improvement formulations [45]. Here ω_t contains model parameters, while Σ_t contains harness code

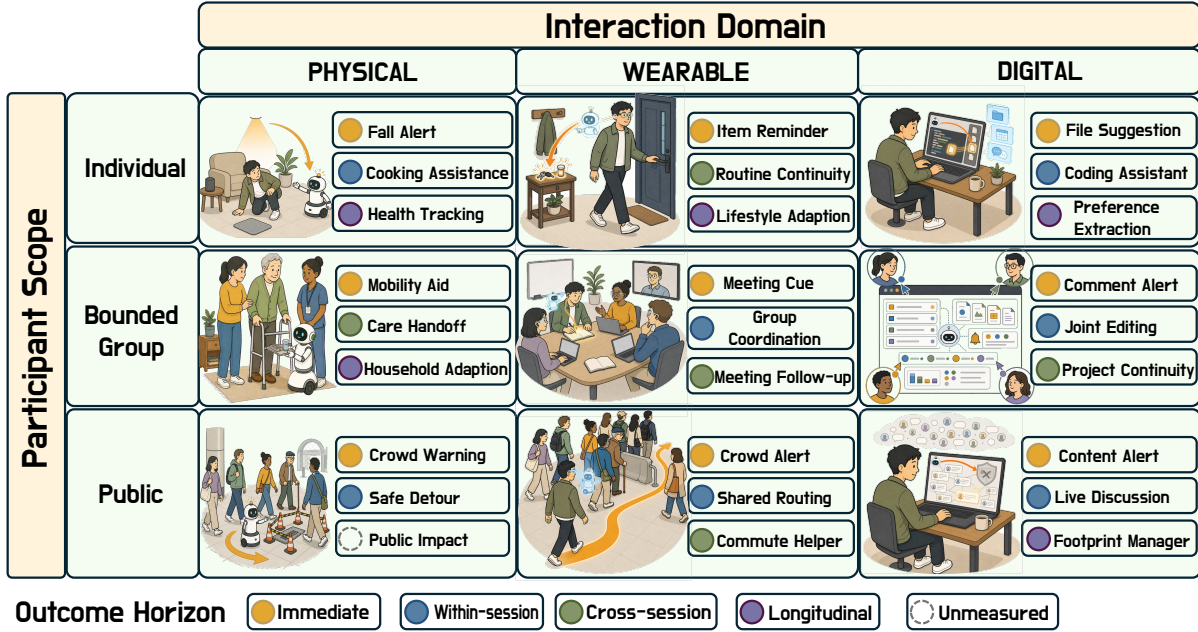


Figure 3: **Three dimensions of application scope:** interaction domain (where observations and actions occur), participant scope (who is involved or affected), and outcome horizon (when consequences become observable). Each cell shows selected examples within that setting.

and configurations implementing these policies, including prompts, tool interfaces, and orchestration logic. Natural-Language Agent Harnesses [83], for example, encode retry, stop, and validation rules in editable documents. Admitted changes to such control rules can revise Σ_t without updating model parameters. By comparison, changing a record of the user’s preferred reminder times updates m_t ; changing the controller’s persistent rule for when to check availability updates Σ_t ; temporarily rephrasing one response commits neither kind of revision. Thus $m_{t+1} \neq m_t \not\Rightarrow \theta_{t+1} \neq \theta_t$.

Classification follows the operational role declared by the implementation. A persistent decision rule stored in a memory database can belong to Σ_t , while a personal fact supplied in a prompt can remain content from m_t . Prompt-level playbooks evolved from execution feedback [84, 85] belong to Σ_t when they govern later decisions. Content updates still need admission gates because accumulated memories and reused unsafe successes can change behavior and safety [86, 87]. The two $A \rightarrow M$ branches therefore share attribution evidence but have different transitions: personal content changes through π_M , while policy revisions pass through evidence construction, proposal, validation, and commit. Neither branch requires the other to update at every opportunity. A stronger recursive case also revises the updater or verifier (Section 8.3).

2.5 Application Scope and Operating Regimes

We characterize application cases by an operating regime $r = (b, n, h)$ along three descriptive dimensions: *interaction domain* b , *participant scope* n , and *outcome horizon* h . Domain identifies the main physical, wearable, or digital setting of observation and action; a service may cross these settings. Participant scope covers the individual, bounded group, or public who are involved or affected. Horizon concerns the particular consequence being evaluated: *immediate*, *within-session*, *cross-session*, or *longitudinal*. The figure marks outcomes without measurements as *unmeasured*. Figure 3 locates illustrative service profiles: fall-alert delivery is (physical, individual, immediate), meeting follow-up is (wearable, group, cross-session), and joint editing is (digital, group, within-session). These examples show the scope of the map without implying deployment or measured outcomes for every profile.

Describing cases through separate properties follows the task-environment perspective in AIMA [88] and provides a common basis for analyzing shared conditions. Interaction domain helps identify resource and sensing constraints, such as wearable energy budgets or uncertainty in physical observation. Participant scope determines whose privacy, consent, and authority must be considered; a meeting assistant may capture information about several people even when its output is private. Outcome horizon affects when evidence becomes available for verification and attributed updates. When feedback arrives across sessions, the user’s circumstances or permissions may have changed. Retaining the context of the earlier decision helps the system interpret that feedback and determine whether it supports a lasting update. These coordinates therefore help explain how the conditions governing local and coupling policies vary across applications.

Always-on describes the continuity of service, while outcome horizon describes the delay to a particular observable consequence. An alert’s delivery may be observed immediately within an ongoing service, whereas the usefulness of a retained preference may emerge across later interactions. The horizon therefore belongs to the evaluated consequence, not permanently to an entire system: observing delivery does not establish an alert’s longer-term benefit.

2.6 Mapping the Literature

Figure 4 maps software and hardware developments onto the framework. Early software foundations address context capture in Cyberguide [89], StartleCam [90], and SenseCam [91]; information organization and retrieval in Memex [92], the Remembrance Agent [93], and Stuff I’ve Seen [94]; and context-sensitive assistance in Lumière [11], CALO [95], and BusyBody [96]. The later module tracks broaden the available spaces and make selection policies and evaluation more explicit. Interface parsing and real-time multimodal processing expand Sensing [97, 98]; managed tiers and revisable records develop Memory [17, 99]; computer-use benchmarks and response activation expose Action’s execution and timing decisions [21, 100]. The Space, Policy, and Benchmark labels distinguish expanded options, decision mechanisms, and evaluation tools.

Hardware changes which signals can be observed, where they are processed, and which effects can be produced. Depth and low-power motion sensors, exemplified by D435i [101] and BMA400 [102], broaden the sensing space. Wearable capture and display platforms connect that space to daily activity [91, 7, 103], while model-equipped handheld and XR interfaces combine perception with digital assistance [104, 105, 106]. Physical platforms further link cameras and onboard processing to manipulation effectors [107, 108]. These changes create shared conditions for the whole service: battery and comfort limit observation, processing location affects privacy and latency, and actuator capabilities constrain feasible actions and recovery.

The tracks converge when current observations, retained context, and timely action must support the same task. PASK [109] connects streaming intent detection, long-term memory, and proactive assistance; AssistantX [6] coordinates perception, planning, decisions, and review for physical collaboration. These systems can be compared by which dependencies they maintain during operation and across interactions. The future direction extends this integration to attributed feedback that can correct personal state and, where justified, support lasting policy revision.

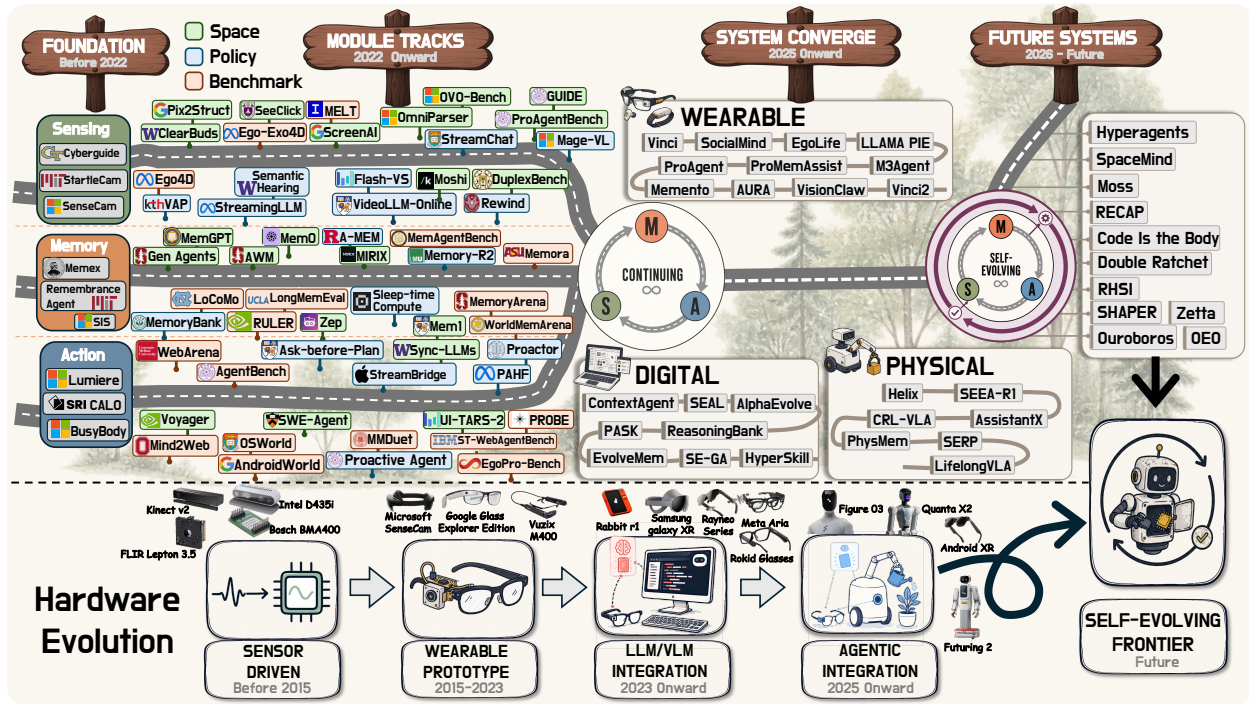


Figure 4: **From module development to system convergence.** Software and hardware advances increasingly connect Sensing, Memory, and Action within personal-AI systems, with governed self-evolution as a future direction.

What Should a Complete Always-On System Include?

Functional requirements. For a declared personal service, a complete system maintains operational continuity and usable durable personal state, realizes the dependencies that service needs, and governs observation, retention, and action under shared conditions. These functional requirements allow shared components and selective channel activation.

Reporting requirements. A report states the service scope, conditions, and outcome horizon, then records the implementation status and supporting evidence for each of the five channels. The mechanisms realizing each dependency and service-specific exclusions are explained alongside those judgments. Complete reporting can document an incomplete system and does not establish service quality.

Self-evolution claims. Governed self-evolution additionally requires outcome attribution, a candidate policy revision, validation, and durable commitment of an accepted change. Reports distinguish these policy changes from personal-content updates; claims of improvement additionally require outcome evidence over the relevant horizon.

3 Sensing

Scheduled and event-triggered sensing supports a continuing loop by creating opportunities to observe between explicit requests. Its contribution to state durability depends on which observations reach Memory and how retained personal context guides later checks. Figure 5 organizes the section around observation streams, operating conditions, local policies, and coupling with Memory and Action. We compare how these choices maintain useful evidence under privacy and resource limits, then examine their evaluation through event

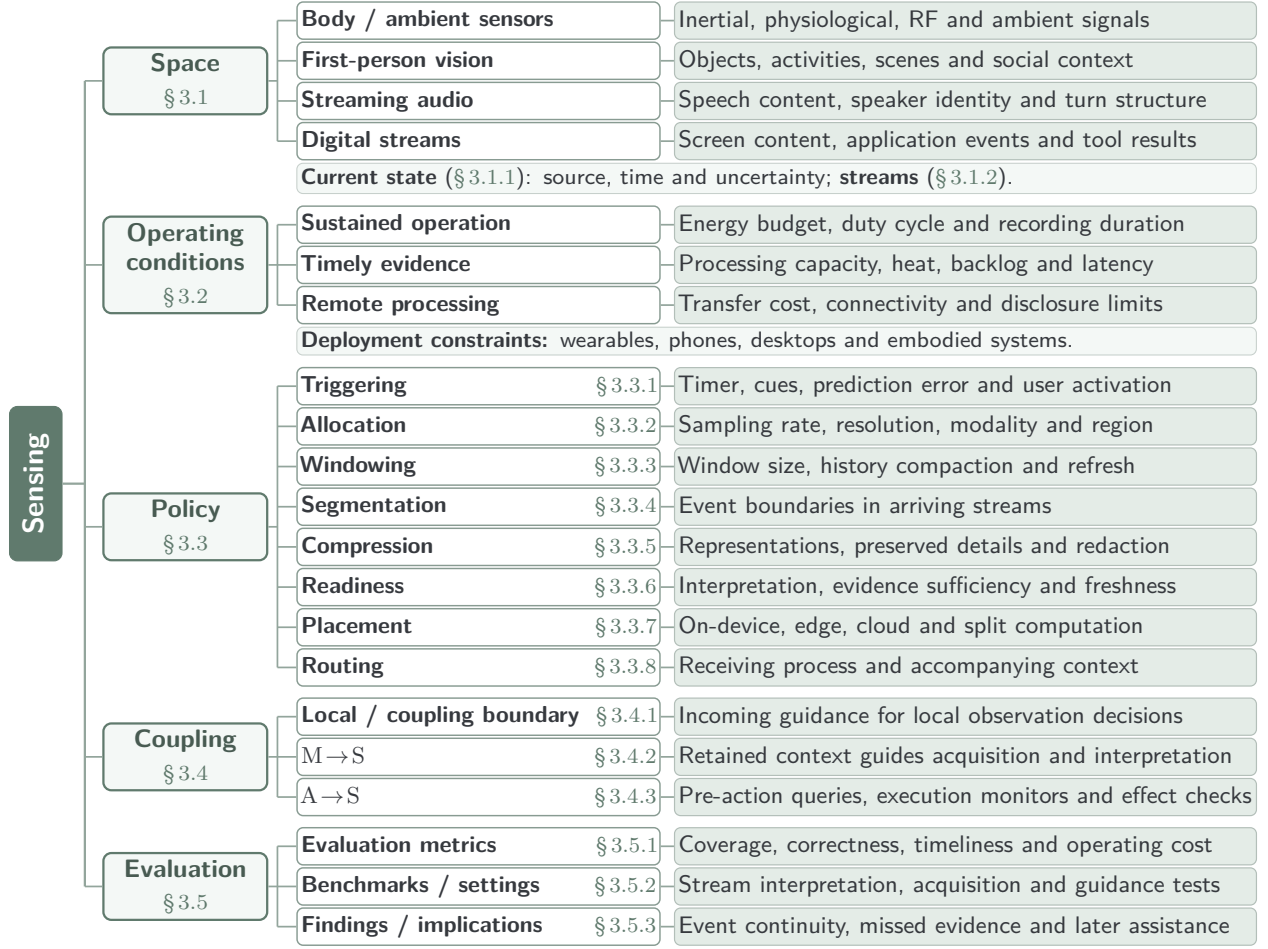


Figure 5: **Sensing section structure and taxonomy.** Observation streams, operating conditions, local policies, incoming $M \rightarrow S$ and $A \rightarrow S$ guidance, and evaluation.

coverage, timeliness, and downstream errors. The self-evolving question is how subsequent outcomes can justify revising the rules for selecting and interpreting evidence.

3.1 Sensing Space

3.1.1 Observations and Current State

The sensing space covers observations available to the system and the current-state representations built from them. Selecting and interpreting observations provides evidence for later decisions [88, 64]. For the reminder assistant, a calendar entry, call-application status, and speech activity can each provide clues about the user’s availability. If the calendar indicates an ongoing meeting while the call application has closed, the state representation needs to preserve source, time, and uncertainty so later decisions can handle the discrepancy. Table 3 summarizes representative sensing methods.

3.1.2 What Can Be Observed

Body and ambient sensors. Inertial, physiological, radio-frequency, and ambient signals can supply task-relevant state directly or provide low-cost cues for richer observation. For example, inertial signals can support gait analysis [110] or help determine when a wearable camera captures images [91]. A continuing service can thus use low-cost activity cues and acquire richer visual context when needed. Keeping these cues useful over repeated wear also requires reliable interpretation as placement and movement patterns

Table 3: **Representative sensing methods and configurations.**

Work	Sensing function	Key feature	Configuration
Body and ambient sensors			
Gait analysis pipeline [110]	Gait estimation	Bouts and foot contacts	Single accelerometer
SenseCam [91]	Capture triggering	Capture on sensor change	Motion / light / infrared
First-person vision			
EgoTrigger [111]	Selective capture	Audio-triggered capture	Raspberry Pi prototype
VideoLLM-online [112]	Stream interpretation	Temporally aligned text	Ego4D / COIN streams
ActionSwitch [113]	Action boundaries	Class-agnostic detection	EPIC-KITCHENS-100
Streaming audio			
Personal VAD [114]	Voice detection	Target-speaker conditioning	Audio + speaker cue
Diart [115]	Speaker attribution	Overlap-aware segmentation	Latency: 0.5–5 s
MOSS Transcribe Diarize [116]	Speech + speakers	Joint timed transcription	Inputs up to 90 min
VibeVoice-ASR-Streaming [117]	Speech + speakers	Incremental attributed text	Checkpoints: ≤ 8 min input
Kirigami [118]	Speech filtering	Remove speech segments	On-device activity sensing
Selective Hearing [119]	Speaker privacy	Bystander-query refusal	Mixed-speaker audio
Digital streams			
ChainStream [120]	Context streams	Process arriving events	Sensor + app inputs
Activity Frames [121]	Activity representation	Timed episodes + sources	Passive desktop capture
A11y-Compressor [122]	Tree compression	Restructure and deduplicate	Accessibility trees

change. Sensor foundation models address related transfer problems across users, devices, and contexts [49]. The information involved also changes with the inference and setting: activity-recognition measurements may reveal wearer identity [123], while smart-home owners and visitors can have different preferences about collection, storage, and sharing [124]. The design must identify needed signals and inferences, and whose information is captured, processed, or retained as participants change.

First-person vision. Egocentric video provides evidence about objects, actions, and social situations surrounding the user. Existing corpora extend this coverage in different ways: Ego4D [125] spans episodic memory, forecasting, manipulation, and social interaction; Ego-Exo4D [126] adds external views; EgoLife [127] extends recording toward multi-day, multi-person activity. These resources support research on recognition and wearable assistance [128, 129, 130]. For a continuing assistant, the relevant event may occur between requests: a brief task-completion event can make a stored reminder obsolete. Mean action durations range from 1.7 to 3.1 s in Assembly101, HD-EPIC, and EPIC-KITCHENS-100 [131, 132, 133], making temporal coverage consequential. Frame rate shapes that coverage, while resolution shapes visible detail; both affect processing cost and the information captured about wearers and bystanders. Evaluations need to specify whether event boundaries and observation windows are supplied or selected online, since this determines whether acquiring timely evidence is part of the task. Capture choices also depend on the social setting: studies of data glasses document context-dependent attitudes toward recording [134, 135] and wearers’ difficulty managing bystander privacy [136]. A camera that follows its user across settings therefore needs capture rules that respond to changes in who is present as well as to the activity being observed.

Streaming audio. Between requests, audio can provide an availability cue or the details of a new commitment. Personal VAD [114] uses the target speaker’s identity to gate on-device recognition, while online diarization adds speaker labels to incoming audio [115]. For a reminder assistant, this distinction helps separate the user’s commitment from another person’s remark. The service may also need to limit which speech enters subsequent processing. Kirigami [118] filters speech locally when only non-speech activity cues are needed, whereas Selective Hearing [119] trains audio models to refuse queries about incidental bystander speech. These choices determine which activity, content, and speaker information remain available as people enter or leave the setting.

The representations also differ in when evidence becomes available. MOSS Transcribe Diarize [116] provides timed, speaker-attributed transcripts for extended-record review, while VibeVoice-ASR-Streaming [117] delivers attributed text as audio arrives. Incremental output can support decisions before a conversation ends. Once the assistant replies, concurrent input and output [98] and capture that continues during decoding [137] keep new evidence available. A correction or change in availability during the reply can then inform whether to continue, pause, or defer. Thus, preserving a conversation for later use and responding to changes during it call for different timing and input arrangements.

Digital streams. Even without a new user request, a continuing assistant needs to detect changes and relate them to prior context. Application events and notifications can trigger observation, while scheduled or on-demand reads recover current state. ChainStream [120] represents sensor readings, screen captures, and messages as streams, processing new items as they arrive. Activity Frames [121] organizes passively captured desktop activity into timed episodes with links to the original evidence. The former supplies context updates, while the latter makes earlier activity available for later interpretation. Background-execution rules, battery use, connectivity, and permissions determine which sources remain accessible. Cross-surface histories extend this scope: MyLifeBits [138] combined personal information sources, and egocentric copilots connect wearable observations to web services [139]. Among the contemporary works inspected, we found no released evaluation linking continuous egocentric and workstation activity for the same person to assistance outcomes.

Available digital inputs include screenshots, document structure, accessibility metadata, and tool responses [21, 140]. Structured inputs expose interface elements directly, while screenshots can supply missing visual information [97]. Their agreement matters when applications change between checks: controlled tests find models following stale structure despite strong image-only accuracy [141]. Accessibility-tree compression and regional summaries reduce the input presented to a model [122, 142]. For recurring observation, compact inputs must retain the details needed to notice changes and resolve conflicts with earlier context.

Tool responses add evidence about execution [143]. Their source and permitted use matter because tool outputs can contain untrusted content [144], and external text can redirect the assistant [145, 146]. Authenticated screens and documents may also contain collaborators’ or correspondents’ information alongside the user’s own content. Policy-constrained task evaluation [147] and region-level privacy judgments [148, 149] examine which information a task permits the assistant to use or disclose. For background reading, selecting relevant regions and preserving their source and audience connects observation to later storage and action permissions.

3.2 Operating Conditions for Continuing Sensing

How long can sensing continue? An always-on assistant needs enough energy to sustain observation under its declared operating schedule, alongside memory access and action [150]. Aria Gen 1 lists 1.5 h of continuous recording under profile 0, with alternative sensor configurations optimized for 5+ h and 15+ h [151, 152]. Lower sampling can extend recording time while leaving short events unobserved, linking battery life to the evidence available for assistance. Figure 6 presents reported power measurements and estimates for representative sensing and processing configurations: wake-up chips consume nanowatt- to microwatt-scale power, while wearable visual processing and phone-based model inference reach milliwatt and watt scales, respectively [153, 154, 155, 156]. This power gap suggests a way to extend battery life: keep low-cost detection available and activate heavier processing when relevant cues arrive.

Can observations reach decisions in time? Continued inference is limited by sustained processing capacity and heat [156]. A test of sustained Qwen2.5-1.5B inference on an iPhone 16 Pro reports throughput settling 41.5% below its peak [157]. When inputs arrive faster than they can be interpreted, a backlog can leave Memory and Action using old context even though capture continues. Acceptable delay also depends on the task: Chen et al. [158] derive target latency ranges of 95–105 ms for continuous pool guidance and 600 ms–2.7 s for step-by-step assistance. A current availability check needs a timely result; a record intended for later review can tolerate more delay. The service therefore needs processing capacity and priorities that match when each observation will be used.

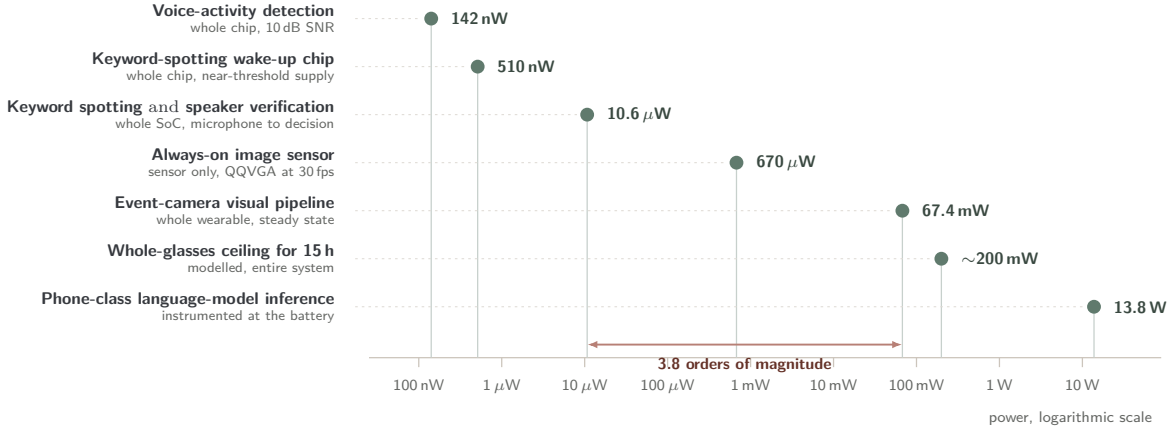


Figure 6: **Published power measurements and estimates.** Values use a logarithmic axis and retain their measurement scope: block, chip, pipeline, or model estimate. The large gaps motivate selective wake-up when a service’s energy budget cannot sustain expensive processing continuously.

What does remote processing cost? Offloading trades local computation for communication. DeepWear [159] shows that transfer overhead can offset faster remote computation. Network delay and disconnection affect when observations become usable. Privacy also constrains what can leave the device, with local filtering providing one way to reduce disclosed content [118]. A continuing service must weigh remote compute against communication cost, connectivity, and permission to transmit evidence.

Why do sensing bottlenecks differ across systems? Wearables follow the user across settings, but small batteries and body-contact thermal limits constrain how long they can maintain sensing [150, 160]. Phones combine sensor and application inputs [120, 161], while background-execution rules and permissions constrain access between requests. Desktops have a larger power budget, but long screen and document streams increase the cost of selecting and interpreting relevant context [122, 142]. For embodied assistants, obtaining a missing view can require physical movement; on a wearable, it may require the user’s effort to look again. Which constraint dominates therefore depends on the application’s required observations and the resources needed to acquire and process them.

3.3 Sensing Policy

Table 4 summarizes eight decisions governed by the local sensing policy π_S , from acquisition to delivery. Under these operating conditions, we compare which events are captured, what information survives processing, and whether it reaches Memory and Action in time to support later decisions. These choices can be combined or revisited as new evidence arrives. A fixed policy can vary the sampling rate with current activity; revising the rule that maps activity to sampling changes the policy used at later opportunities. This separates operating a continuing service from revising its observation rules.

3.3.1 Whether to Observe: Triggering and Wake-Up

Scheduled and user-initiated capture. SenseCam [91] supports timer-based photographs, a manual shutter, and start/stop recording. Scheduled capture builds background context without repeated user intervention; manual control lets the user select a moment or recording period but requires their attention. The appropriate interval depends on the scene: ProAgent [179] finds that low visual sampling rates have less impact in stationary settings but can miss critical events in dynamic scenes.

Cue-based activation. Triggers act at different points in acquisition and processing. EgoTrigger [111] classifies hand–object interaction sounds to trigger image capture. OpenGlass [155] uses low-power wake-up sensing to activate a separate inference processor that is powered down between inferences. ProAgent [179]

Table 4: **Local Sensing decisions and representative works.**

Policy	Main choices	Condition and cost	Representative works
Triggering	Timer / cue / prediction error / user	Sparse events: missed activation / checking cost	SenseCam [91], EgoTrigger [111], Cambrian-S [162]
Allocation	Rate / resolution / modality / region	Brief events: temporal coverage / detail per frame	Q-Frame [163], AwaRes [164], Ego-METAS [165]
Windowing	Window size / compaction / refresh	Long streams: useful context / processing cost	StreamingVLM [166], JoyAI [167], StreamChat [168]
Segmentation	Onset / offset / overlap / change	Arriving streams: stable boundaries / delay	StartNet [169], ActionSwitch [113], On-GEBD [170]
Compression	Select / summarize / extract / filter	Future use unknown: retained detail / disclosure	Imprint [171], Kirigami [118]
Readiness	Proceed / wait / observe more	Uncertain evidence: premature use / waiting	SmolVLA [172], DynamicVLA [173], N ₀ -TWAM [174]
Placement	Sensor / device / edge / cloud	Limited device budget: compute / transfer and disclosure	EgoCHARM [175], DeepWear [159], Neurosurgeon [176]
Routing	Decisions / records / background	Multiple recipients: timely delivery / coordination	Dispider [177], MIRIX [18], Thinking Machines [178]

The condition/cost column states the decision pressure and trade-off; it does not rank the cited implementations.

combines regular sampling with motion, location, and conversation cues that increase the visual sampling rate. After acquisition, PRPF [180] uses a lightweight multimodal perceptor to gate a larger reasoner; StreamMind [181] places a learned event gate between its video encoder and language model. These choices avoid different costs: image capture, processor operation, or model invocation.

Prediction-error triggers. Cambrian-S [162] predicts latent frame representations and uses discrepancies to identify boundaries and trigger interpretation of accumulated video. Across trigger designs, savings depend on the checking cost and activation frequency. False triggers waste capture or computation; missed triggers can leave decisions using outdated context when no other source records the change.

3.3.2 How Much to Observe: Allocating the Perception Budget

Sampling rate and representation. VideoLLM-online [112] pools each frame into one token, while StreamMind [181] combines spatial and temporal features in one perception token per frame. For a fixed representation, visual tokens per second equal frames per second multiplied by tokens per frame. Spending more tokens on each frame therefore leaves room for fewer frames under the same budget [182]. Apollo’s training ablations [183] favor fixed-rate sampling over a fixed frame count, while F-16 [184] combines 16-fps input with within-clip compression to retain fast motion. In our calculation from HD-EPIC’s released annotations [132], 40.6% of action intervals are shorter than one second,¹ so brief events can fall entirely between successive frames sampled uniformly at 1 fps.

Frame selection and reuse. AKS [185] selects keyframes under a fixed budget, while LVNet [186] uses hierarchical selection to reduce long inputs. Q-Frame [163] jointly selects moments and resolution according to the query. VisualClaw [187] compares selective and uniform sampling at eight frames per query, isolating frame choice within the same budget. STC [188] instead reuses and prunes similar visual information during streaming encoding, reducing repeated processing of observations that have already arrived.

Resolution and regions. VisionThink [189] starts with a low-resolution image and requests higher resolution when needed; AwaRes [164] confines added detail to relevant crops while retaining a low-resolution

¹Duration statistics use `analysis/reproducibility/action_duration_stats.py` in the accompanying repository; the paired JSON records source versions, filters, and counts.

global view. CropVLM [190] similarly supplies a relevant crop alongside the full image, with an ablation showing reduced performance when the full image is removed. HyperVL [191] chooses resolution before visual encoding and reports encoder latency and memory on an edge platform. The comparison separates view-guided refinement from advance allocation and shows why local detail may require wider context.

Modalities and task requirements. Ego-METAS [165] evaluates per-step modality selection under hardware-based energy budgets on continuous egocentric streams. Useful sensor combinations depend on the scenario: learned routing struggles on untrimmed streams, while simple dynamic fusion offers useful accuracy–energy tradeoffs. Task representation also matters. Zhao et al. [192] improve spatial generalization by omitting proprioceptive policy inputs when actions use relative end-effector motions and cameras preserve task-relevant coverage. Modality selection thus depends on both acquisition cost and the information needed by the downstream task.

3.3.3 How Long to Observe: Observation Windows

Recent context and compacted history. Observation windows bound the history available to interpret the current situation. StreamingLLM [193] retains initial and recent tokens in a bounded cache; StreamingVLM [166] uses shorter visual and longer text windows. Flash-VStream [194] preserves detailed recent frames and compacts older observations into temporal and abstract visual representations. JoyAI-VL-Interaction [167] retains recent visual tokens, summarizes earlier chunks as text, and consolidates older summaries at a coarser level. Window limits discard history, while compaction preserves a less detailed version. The choice controls which earlier instructions and corrections accompany current activity.

Updating the window during a response. StreamChat [168] refreshes visual context at each decoding step, allowing new frames to affect an ongoing answer. Audio-Interaction [137] continuously encodes audio into a queue but incorporates it at designated points, such as silence or the end of a response; during generation, its decoder uses context already received. Thus, history length and refresh timing are separate choices. A spoken correction may already be captured yet still await entry into the model’s context.

3.3.4 Where to Cut: Online Segmentation

Action progress and onsets. Streaming temporal action segmentation labels arriving video chunks without future observations [195, 196]. OnlineTAS [197] uses adaptive context and post-processing to reduce fragmented predictions; ProTAS [198] adds action progress and learned task graphs to locate procedural steps. Action-start detection [199, 200, 169] instead locates onset, enabling intervention before completion. For bounded history processing, LSTR [201] combines coarse historical context with fine recent detail, whereas TeSTra [202] reuses temporal attention computations to keep each update at constant cost.

Overlapping and unfamiliar events. ActionSwitch [113] detects simultaneous action intervals without class labels. Object-aware egocentric detection [203] uses the objects involved, while Mamba-OTR [204] targets sparse take-and-release events. Generic event boundary detection [205] locates meaningful changes without naming actions; UBoCo [206] learns boundaries without manual boundary supervision, and On-GEED [170] decides online. These methods yield different units: concurrent activities, object transitions, or broader changes that prompt reinterpretation.

Boundaries from prediction and change. Event segmentation theory links boundaries to discrepancies between expected and observed events [207, 208]. Cambrian-S [162] uses latent prediction error, SUNTA [209] studies surprise-based chunking for hierarchical video prediction, including inconsistency during imagined rollouts, and Event-VStream [210] combines motion, semantic, and predictive cues to detect transitions and trigger language generation. ES-Mem [211] applies event segmentation to dialogue memory. Boundaries shape when processing resumes and which observations form a unit for later retrieval.

Boundary timing and service value. StartNet [169] evaluates action starts with offset tolerances from one to ten seconds. Fine-grained actions average 1.7 s in Assembly101 [131], 2.0 s in HD-EPIC [132], and 3.1 s in EPIC-KITCHENS-100 [133]; our calculation from the released annotations gives medians of 1.18 s and

1.62s for the latter two datasets. A tolerance longer than an event can credit a prediction arriving after it ends. Such a prediction may support retrospective recording but arrive too late for intervention, so boundary accuracy needs to be interpreted alongside detection delay.

3.3.5 What to Keep: Compression, Representation, and Redaction

Visual evidence and spatial reuse. Retaining selected observations preserves details for later inspection. StreamVLN [212] combines a recent dialogue window with slower visual memory and prunes repeated observations through 3D voxel correspondence. Its spatial-pruning variants require depth and pose beyond the RGB-only base. This removes repeated views of the same scene content, while the remaining visual detail determines what can later be checked, including changes that isolated salient frames may omit.

Aligned streams and structured records. LiveCC [213] interleaves video frames and transcript words at their timestamps, preserving temporal alignment during training. Imprint [171] extracts person–action–object records with time intervals and consolidates them using recurrence, recency, and distinctiveness; compared with hierarchical caption summaries, it improves long-horizon question answering while reducing storage and retrieval cost. These representations retain different structure: word–frame timing in the aligned stream and recurring relationships in event records. SWE-Pruner [214] instead filters file lines by the agent’s information request, returning the full output if none is supplied.

Filtering sensitive content. Kirigami [118] filters human speech locally while preserving audio for activity recognition, evaluating reduced speech recoverability together with remaining recognition accuracy. The appropriate tradeoff depends on the task: availability detection can use non-speech cues, while retaining a spoken commitment requires its words. Filtering before transmission or storage changes disclosure and future answerability; representation choices therefore include where and what to redact.

3.3.6 Whether Enough Has Been Observed: Interpretation and Readiness

Evidence sufficiency and response timing. Readiness determines whether available evidence supports the next decision or whether the assistant should wait or observe more. A query can arrive before its answer is observable; an apparently sufficient observation can also become stale during inference. Even when a call is known to have ended, Action may defer a reminder because the timing or permission is unsuitable.

Freshness during execution. SmolVLA [172] overlaps execution with fresh observation and inference before its action queue is exhausted. DynamicVLA [173] aligns queued actions with elapsed inference time to reduce outdated execution. FASTER [215] prioritizes near-term actions during sampling so movement can begin before the full action chunk is ready. The mechanisms address different delays: obtaining the next observation, aligning execution with it, and releasing a usable result.

Task-specific confirmation. In contact-rich manipulation, N_0 -VTLA [216] conditions control on measured touch and predicted tactile change. N_0 -TWAM [174] additionally uses observed contact events to advance between task stages, such as after a completed grasp. Prediction prepares control, while an observed transition confirms that the procedure can continue. This distinction matters when a plausible interpretation or an elapsed interval does not establish that the required event occurred.

3.3.7 Where the Computation Runs: On-Device, Edge, and Cloud

Local processing and computation splits. EgoCHARM [175] performs inertial activity recognition on an IMU coprocessor. Split inference assigns part of the workload elsewhere. Neurosurgeon [176] runs initial layers on a mobile device and remaining layers in the cloud, selecting the split using computation costs, intermediate-data sizes, network conditions, and server load. DeepWear [159] partitions inference between a wearable and its paired phone over Bluetooth, balancing latency and energy without Internet access.

Choosing what crosses the link. Placement also determines which representation leaves the device. Shredder [217] adds learned noise to intermediate features before cloud inference and compares information

leakage with inference accuracy. Choosing a split point and protecting the transmitted representation are separate decisions: keeping computation local avoids that transfer, while feature protection changes the information available remotely. A useful split must return task-relevant evidence within the service’s latency, energy, and privacy constraints.

3.3.8 Where the Result Goes: Routing

Different consumers require different evidence. Dispider [177] supplies compressed clip features to a lightweight decision process and retrieves historical clips for a separate response module; monitoring continues while a response is generated. MIRIX [18] combines incoming content with retrieved context, then uses a central memory manager to select specialized managers for updates. The former separates evidence for immediate control from evidence for response generation; the latter separates choosing a persistent destination from maintaining its records.

Background work also needs context that remains interpretable when a delayed result returns. Thinking Machines’ interaction-model architecture [178] passes conversational context to asynchronous reasoning while the interaction model continues receiving input. Returned results must be reconciled with intervening changes, such as cancellation. Routing therefore selects both the receiving process and the context accompanying the evidence: a transient availability cue may support only the current decision, while a new commitment needs a record whose source and timing support later correction.

3.4 Sensing Coupling Across the Loop

The local policies above determine how observations are acquired, interpreted, and delivered. In a continuing loop, these decisions also depend on what the assistant remembers and what it is trying to do. We examine the two incoming channels, $M \rightarrow S$ and $A \rightarrow S$, through the information they supply, the sensing decisions they guide, and the conditions under which that guidance helps or fails. Table 6 compares representative implementations by their guidance, sensing use, and reported evaluation.

3.4.1 Local Decisions and Coupling Boundaries

Table 5 illustrates the different sources of guidance for sensing decisions. Local processing in Sensing covers the acquisition and interpretation of observations. Examples include triggering capture from motion in the current stream and using a short frame buffer to interpret the scene. $M \rightarrow S$ is involved when retained personal state guides acquisition or interpretation, such as using a persistent identity record to recognize a person. $A \rightarrow S$ is involved when an intended action, plan, or execution state supplies an information need or observation target. These channels guide the local policy, which resolves the sensing choice under current inputs and shared conditions. Both can influence the same decision.

A remembered routine can suggest when to check availability, while a pending reminder specifies what needs confirmation. Personal context may first help the assistant decide what to ask, with the selected question determining the next observation. Feedback can correct the stored routine and thereby change later checks under the same coupling rules. Self-evolution can instead revise those rules: how old a routine may be before $M \rightarrow S$ stops using it as observation guidance, or how $A \rightarrow S$ turns execution requirements into checks for future tasks. The analysis therefore distinguishes a change in the supplied prior or target from a lasting change in how such guidance is produced.

Table 5: **Local sensing decisions and incoming guidance.**

Relation	Description	Example
Local policy	Makes acquisition and interpretation choices	Current motion triggers image capture
$M \rightarrow S$	Retained personal state guides Sensing	A stored identity helps recognize a person
$A \rightarrow S$	Action context supplies observation targets	A planned grasp prompts a position check

3.4.2 Memory to Sensing: the $M \rightarrow S$ Channel

Choosing further observations with retained context. Retained context can guide which evidence the assistant acquires next. ProAgent [179] retrieves scenario-relevant personas for its assistance-need decision, then uses that decision to maintain or reduce subsequent visual sampling. Removing this feedback to sampling lowers proactive F1, and its sampling comparisons examine coverage alongside the fraction of frames acquired. Retained personas first guide Action, and the resulting need estimate then regulates Sensing. VisionClaw [10] illustrates a digital counterpart. Personal email or calendar context supplies targets for delegated searches, such as using a colleague’s name to initiate a sequence of web lookups. Its user studies compare the combined service with separate perception and agent conditions. ProAgent concentrates observation over time, while VisionClaw directs it toward relevant external sources. A persona alone may not specify a useful sampling rate or search target. Relating it to the assistance being considered supplies that missing task context. This can focus acquisition on evidence relevant to the service, while making it depend on the accuracy of the assistance decision. A mistaken need estimate can reduce sampling too early, and a mistaken referent can send a background search to the wrong person or topic.

Interpreting new observations with retained context. Once observations arrive, retained context can help resolve their ambiguity. M3-Agent [130] matches faces and voices in incoming clips to persistent identity nodes before producing new event descriptions. It also accumulates evidence for face–voice equivalences, allowing repeated consistent associations to outweigh occasional conflicting links. Its identity-equivalence ablation lowers later question-answering accuracy, connecting this representation choice to the usefulness of the resulting records. SpeechLess [218] uses personal spatial episodes to infer missing parts of a short utterance or context-only request. Its interaction study finds reduced articulation effort, with additional inference delay and lower accuracy for some abbreviated modes. All modes share the retriever, so this comparison concerns the interaction supported by memory. Together, the methods illustrate two benefits: stable references across observations and less repeated explanation by the user. They also expose different risks. A wrong identity match can propagate to later records, while an overly specific interpretation can make an ambiguous request appear settled before the user has confirmed it.

Identity guidance can also operate before language interpretation. Personal VAD [114] uses an enrolled voice embedding stored on the device to detect the target speaker. VoiceFilter-Lite [219] uses the same kind of personal reference to suppress competing speech before recognition. The former controls which speech activates processing, while the latter controls what remains in the signal. Their detection and recognition comparisons address different points at which another person’s speech can enter a continuing assistant’s evidence. Memento [220] applies retained references to visual localization: an object named in a retrieved memory becomes a detector target in the current view. These mechanisms connect persistent identity or object references to current evidence without requiring the user to name them again.

Why memory-guided sensing is difficult. Retained state influences both the interpretation of evidence and the opportunity to obtain counterevidence. After an interpretation error, a record may still contain enough observations to identify the mistaken association. After a sampling error, the relevant change may never enter the record. A prediction-error trigger cannot detect a change in a stream that is no longer being sampled. PAHF [221] shows how this problem extends to clarification, using simulated users whose preferences change. Its pre-action-only configuration asks questions when retrieved preferences leave ambiguity. Once the agent has learned confident but subsequently outdated notes, it asks few further questions and adapts poorly to the changed preferences. Configurations with post-action feedback can correct those notes even when the agent no longer perceives a need to ask. Here memory influences observation by changing whether Action asks a question. Clarification can prevent an initial mistake, while later user correction can expose a confidently held belief that has become wrong. In continuing assistance, reducing questions or background checks can save effort while also reducing opportunities to notice that the user’s needs have changed.

Keeping those opportunities available requires mechanisms for revisiting earlier assumptions. SLiM [222] maintains uncertain beliefs about target and landmark locations, using their spatial relations to choose useful views. Its search comparisons show why treating a noisy landmark location as fixed can mislead exploration. Active Neural Mapping at Scale [223] uses uncertainty in an evolving map to guide further views while

accounting for traversable paths and nearby regions. Its ablations distinguish selecting uncertain locations from organizing an efficient route between them. These spatial-belief and mapping studies suggest ways to maintain personal environmental knowledge: an old object location can prioritize an initial check, while uncertainty and a failed observation justify looking elsewhere. Each approach has a cost. Revisiting a location may require movement, renewed access, or user cooperation, while more frequent checks increase acquisition effort. The design problem is therefore to retain the savings from personal guidance while leaving enough evidence available to correct it as routines, identities, and permissions change. A prior can suppress its own corrective evidence, motivating the qualitative High rating in Table 2.

3.4.3 Action to Sensing: the $A \rightarrow S$ Channel

Obtaining information before commitment. An intended action gives the assistant a reason to obtain particular evidence. Before a purchase, the missing information may be a preference that changes which option is acceptable. Before a reminder, it may be whether the user is still occupied. Ask-before-Plan [224] combines user constraints with observations from tool calls to identify missing or infeasible requirements and ask targeted clarification questions before producing a plan. SpeechLess [218] provides a more constrained interaction: a low-confidence response is presented with a rationale and a yes/no verification before it is stored. KnowNo [225] adds a calibrated choice: it forms a set of plausible actions and requests human help when the set contains multiple options. Its simulated and real-robot comparisons examine task success together with the frequency of help, making the cost of resolving uncertainty explicit. Open clarification can uncover an unanticipated requirement, while confirmation makes an already proposed interpretation easier to accept or reject. These choices trade information breadth against user effort and delay. In a continuing service, a question should resolve uncertainty relevant to the pending decision, and its usefulness depends on whether the user can answer before the opportunity to help passes.

Predictive models can supply expected consequences that help identify what remains uncertain before acting [79, 65]. OmniManip [226] checks rendered constraints before execution and uses object-pose feedback during motion. ICWM [227] acquires probing interactions and tests their contribution to later control. Questions, constraint checks, and object probes supply different evidence for the pending choice.

Monitoring conditions during execution. Some requirements must hold throughout an action. DoReMi [228] derives constraints from the current plan and uses a vision-language model to check them during skill execution. Detected violations return to the planner for recovery. Code-as-Monitor [229] makes this process more explicit by translating task constraints into geometric elements and executable monitoring code. It tracks the relevant points, lines, or surfaces, avoiding repeated full VLM calls for each check. Its ablations show that constraint-specific elements and multiple views contribute to success. The comparison reveals a tradeoff between flexible semantic checks and efficient, precisely grounded monitors. DoReMi reports difficulties with occlusion and visual mis-detection. Code-as-Monitor’s compact geometry can miss failures involving force without visible displacement or omitted cues such as flowing water and sound. A persistent monitor therefore needs an observation representation that contains the failure signal, not merely an inexpensive test of the chosen constraint. The same issue arises when a personal assistant monitors whether an ongoing background task still satisfies the user’s instructions.

Plan, Watch, Recover (PWR) [230] extends plan guidance to the observation window. The planner supplies visual cues for step progress and completion, while plan-update timestamps anchor clip selection. Its plan-condition comparisons examine intervention timing and procedural assistance. For continuing tasks, this couples what to look for with which earlier moments to revisit, so a bounded observation window can retain relevant procedural context.

Checking effects after execution. Digital interfaces make the distinction between issuing an action and observing its result especially clear. VeriGUI [231] records an expected effect, checks subsequent visual evidence against it, and incorporates verification into corrective action selection. Its comparisons report improvements in execution-error recovery, while its robustness benchmark primarily simulates failed operations that leave the screen unchanged. VisCritic [232] instead learns an action-aware comparison of pre- and post-action screenshots. Its interactive comparisons examine visual versus textual critics, with limitations arising

from subtle text changes, animations, and asynchronous interface updates. These approaches emphasize different parts of the dependency: specifying the expected transition and recognizing the transition that occurred. Together they show why an unchanged screen can indicate a missed click, but a changed screen can still reflect the wrong navigation or an unrelated update. For a continuing desktop assistant, the next step depends on identifying which change matters to the task, including whether a result has completed, remains pending, or needs a different recovery action.

When a check leaves uncertainty unresolved, the next observation becomes part of recovery. VAGEN [233] progressively acquires evidence through tools. NavMCP [234] varies observations and cross-call state within navigation episodes that reset exploration and memory. ExeVRM [235] studies a different part of this process: assessing success and locating errors from a supplied execution video and task instruction. Its evaluation concerns interpretation of recorded results. These settings distinguish learning that an operation failed, locating the error, and acquiring evidence for the next attempt.

Why action-guided sensing is difficult. An action supplies a purpose for observation, but the available source must provide evidence that can answer the corresponding question. A tool receipt may confirm that a command was accepted, whereas visible completion or a side effect may require inspecting the resulting state. Interactive perception [69] studies actions that reveal otherwise inaccessible properties, making acquisition itself part of the task. Calandra et al. [236] show a complementary benefit from combining vision with touch during grasping and regrasping, particularly when visual appearance alone gives weak evidence about contact. Their model uses manipulation feedback to select the next grasp. Probing an object, inspecting an existing result, and consuming a tool acknowledgment consequently answer different questions at different costs. For personal assistance, asking the user to look again adds effort, moving a robot adds physical cost, and opening another page requires access. The coupling policy identifies the unresolved condition, while the local sensing policy selects a feasible source under the current resource and permission constraints.

Even a suitable source can be inconclusive when observed at the wrong time. A missing result may reflect failure, delay, or a view that hides the effect. Foresight [237] studies this ambiguity through action-conditioned world-model representations: the same object on a table can be expected before a grasp, indicate a missed lift, or confirm a completed placement. It learns failure detection from final task-level labels, illustrating the challenge of locating failure onset without dense temporal supervision. For continuing service, the monitor must also follow changes in the plan and stop checks that a cancelled request has made obsolete. Checking too early can trigger a needless retry, while checking too late can allow subsequent actions to depend on a false assumption. Useful $A \rightarrow S$ guidance therefore keeps observations aligned with the active task and the time at which its expected effects can be assessed. Table 2 assigns a qualitative Moderate rating: the action usually specifies the information need, while obtaining timely evidence remains the main challenge.

Table 6: **Representative implementations of $M \rightarrow S$ and $A \rightarrow S$.** Rows connect supplied guidance to its sensing use and the original study’s evaluation. * marks memory guidance implemented through an action decision.

Work	Guidance supplied	Use in Sensing	Reported evaluation
M $\rightarrow$ S: retained state guides observation and interpretation			
ProAgent [179]*	Retained personas	Adjust sampling via Action	Sampling/loop ablations
VisionClaw [10]*	Personal calendar/email	Direct external searches	Combined-system study
ProAct [238]*	Memory gaps and goals	Search missing subtopics	Directed-search ablation
Personal VAD [114]	Enrolled voice embedding	Detect target-speaker speech	Detection precision–recall
VoiceFilter-Lite [219]	Enrolled voice embedding	Suppress competing speech	Recognition error/noise
SocialMind [239]	Stored face identities	Identify known partners	Persona-module ablation
M3-Agent [130]	Face/voice identity nodes	Ground new event records	Identity-link ablation
Memento [220]	Stored memory referents	Locate objects in view	Multi-day usage study
AiGet [240]	User interests/expertise	Interpret scene familiarity	Profile/rule ablations
SpeechLess [218]	Personal spatial episodes	Resolve partial intent	Speech-mode comparison
IPI-Agent [241]	Retained monitoring tasks	Match task cues to frames	Control/gating ablations
A $\rightarrow$ S: action context supplies information needs and observation targets			
KnowNo [225]	Uncertain action options	Ask user to resolve options	Success/help-rate tradeoff
OmniManip [226]	Manipulation constraints	Pre-/in-action checks	Manipulation tasks
ICWM [227]	Unknown interaction state	Probe before control	Probing contribution
Ask-before-Plan [224]	Missing/infeasible constraints	Ask targeted questions	Clarification and plan quality
Inner Monologue [242]	Planner information needs	Request missing scene facts	Feedback-source comparison
ContextAgent [243]	Planned tool dependencies	Read location, then weather	Tool-selection accuracy
Alpha-Service [244]	Dispatched service tasks	Set visual triggers/queries	Service demonstrations
PWR [230]	Step cues and timestamps	Select clips and track steps	Plan-condition comparison
DoReMi [228]	Plan constraints	Monitor visual violations	Detection and task success
Code-as-Monitor [229]	Geometric task constraints	Track points, lines, surfaces	Geometry/view ablations
AssistantX [6]	Expected task outcomes	Check messages/robot state	Agent-module ablations
VeriGUI [231]	Expected interface effects	Check post-action screens	Execution-error recovery
VisCritic [232]	Executed UI action	Compare before/after views	Visual/text critic comparison
VAGEN [233]	Unresolved task state	Acquire more evidence	Progressive tool checks
NavMCP [234]	Navigation progress	Update the next observation	Within-episode comparisons
Foresight [237]	Robot action sequence	Interpret visual failure cues	Latent-feature ablation
PITCH [245]	Earlier daily plan	Ask about completion later	Daily-dialogue analysis

3.5 Evaluation

3.5.1 Evaluation Metrics

Coverage and correctness describe which events are captured and how faithfully they are interpreted, using event recall, false-trigger rates, recognition accuracy, and segment-level scores. Measuring omissions requires an event reference independent of the selected observations. Timeliness measures when sufficient evidence becomes available and when a response follows. Latency comparisons need explicit start and end points, including whether capture, processing, and transmission are counted. Operating cost includes sensor activity, computation, and communication. Evaluating quality and delay alongside these costs exposes the tradeoffs relevant to maintaining observation between requests.

3.5.2 Benchmarks and Evaluation Settings

Table 7 groups evaluations by Sensing space, local policy, and coupling policy. With a supplied video stream, OnlineTAS [197] measures both frame accuracy and event segmentation. On 50Salads, its online post-processing raises the edit score from 28.8 to 71.5 while frame accuracy falls from 80.9% to 77.7%, exposing the difference between frame recognition and event continuity. ProReady-QA [246] adds a different temporal requirement: annotated answer-evidence windows let correctness be scored together with premature and delayed responses. These protocols examine the interpretation and timely use of evidence already supplied.

Acquisition comparisons let the policy choose which evidence becomes available. EgoTrigger [111] compares full capture, uniform subsampling, and audio-triggered capture on QA-Ego4D and HME-QA. Its fixed-duration trigger retains comparable memory-QA performance with roughly half as many frames as full capture. Its prototype power measurements additionally allow the input reduction and the measured hardware energy result to be considered separately. Ego-METAS [165] compares sensor selection for online segmentation under energy budgets. Both can test observation choices using recorded streams, with selected inputs and acquisition costs represented in the evaluation. This differs from simply processing a supplied stream chronologically. For evidence already acquired, a matched frame budget makes SimpleStream’s recent-frame baseline [247] a useful control for more elaborate history selection.

Guidance comparisons test whether supplied context improves the receiving decision. PWR [230] compares absent, generated, and oracle plans, and reports differing rankings for response quality and intervention timing across plan conditions. Informative guidance and its timely use therefore need separate outcomes, even when monitoring and response share an implementation.

Table 7: **Sensing benchmarks and evaluation protocols.** Groups indicate the primary sensing question. Rows labelled *tests* refer to evaluations within the cited work.

Benchmark / protocol	Space / policy target	Evaluated content
Sensing space: what can be observed		
ExtraSensory [248]	Body/ambient signals	Daily context from phone/watch sensors
Ego4D [125]	Egocentric vision	Activity, recall and social understanding
Ego-Exo4D [126]	Egocentric vision	Skilled activities across viewpoints
EgoLife [127]	Egocentric vision	Multi-day event recall and reasoning
WearVQA [249]	Egocentric vision	VQA under authentic capture conditions
WearVox [250]	Streaming audio	Noise robustness and side-talk rejection
ScreenSpot-Pro [251]	Digital streams	GUI grounding on high-resolution screens
WebPII [149]	Digital streams	Sensitive-region detection in screenshots
Local sensing policy: acquiring and processing evidence		
Speech Commands [252]	Triggering	Keyword detection on supplied audio
EgoTrigger tests [111]	Triggering	Capture policies: frame count and QA
Ego-METAS [165]	Budget allocation	Sensor selection under energy budgets
Kinetics-GEBD [205]	Segmentation	Event boundaries without action classes
OnlineTAS tests [197]	Segmentation	Frame/segment scores on supplied video
VStream-QA [194]	Observation window	QA over a supplied video stream
SimpleStream tests [247]	Observation window	Recent-frame budget versus history use
TeleEgo [253]	Window / readiness	Streaming recall, accuracy and timeliness
OVO-Bench [254]	Window / readiness	Supplied video prefix: recall and response
EgoSAT [255]	Readiness	Answerability from observed frames
ProReady-QA [246]	Readiness	Correctness within answer-evidence windows
ESTP-Bench [256]	Readiness	Timely responses to evolving questions
EgoPro-Bench [257]	Readiness	Intent recognition and interaction timing
Full-Duplex-Bench [258]	Readiness / turn timing	Pauses, interruptions and turn-taking
Coupling policy: retained state and action guidance		
Personal VAD tests [114]	M → S: speaker detection	Detection with enrolled-speaker guidance
VoiceFilter-Lite tests [219]	M → S: speaker filtering	Recognition with competing speakers
IPIBench [241]	M → S: task matching	Monitoring tasks added, edited or cancelled
PWR plan tests [230]	A → S: step monitoring	Intervention timing across plan conditions
SentinelBench [259]	A → S: monitoring targets	Condition-based waiting: delay and cost

3.5.3 Findings and Implications

Tracking activities benefits from evaluating event continuity alongside frame accuracy. Time-sensitive assistance needs responses aligned with evidence windows. For memory assistance, selective capture has preserved measured QA utility at lower input cost. Extending these findings to continuing service requires

examining events relevant to other commitments and later interactions. An incorrect event label leaves an output to inspect, whereas an uncaptured event can escape answer-based scoring entirely. For a self-evolving loop, this also constrains the evidence for revising sampling, compression, or the rules by which Memory and Action guide observation. A revision that reduces processing cost must still be checked for missed events that change later assistance, including events the previous policy did not select.

Selective Observation Can Limit Later Correction. Sensing must supply evidence in time for decisions or later review, within power, latency, and privacy constraints. Memory and Action can focus acquisition, but stale or narrow guidance can leave corrective evidence unobserved. Action-guided checks further depend on whether the chosen source and timing can address the relevant uncertainty.

4 Memory

Memory provides the continuing loop with usable personal and task state across interactions. This state durability supports resumed assistance, while maintenance must keep relevant records available and carry corrections into later decisions. Figure 7 organizes the section around memory space, local maintenance policies, coupling, and evaluation. We compare how retained information gains, maintains, and eventually ceases to have influence. For the self-evolving extension, we distinguish updating that information from revising the rules governing its admission, use, and correction.

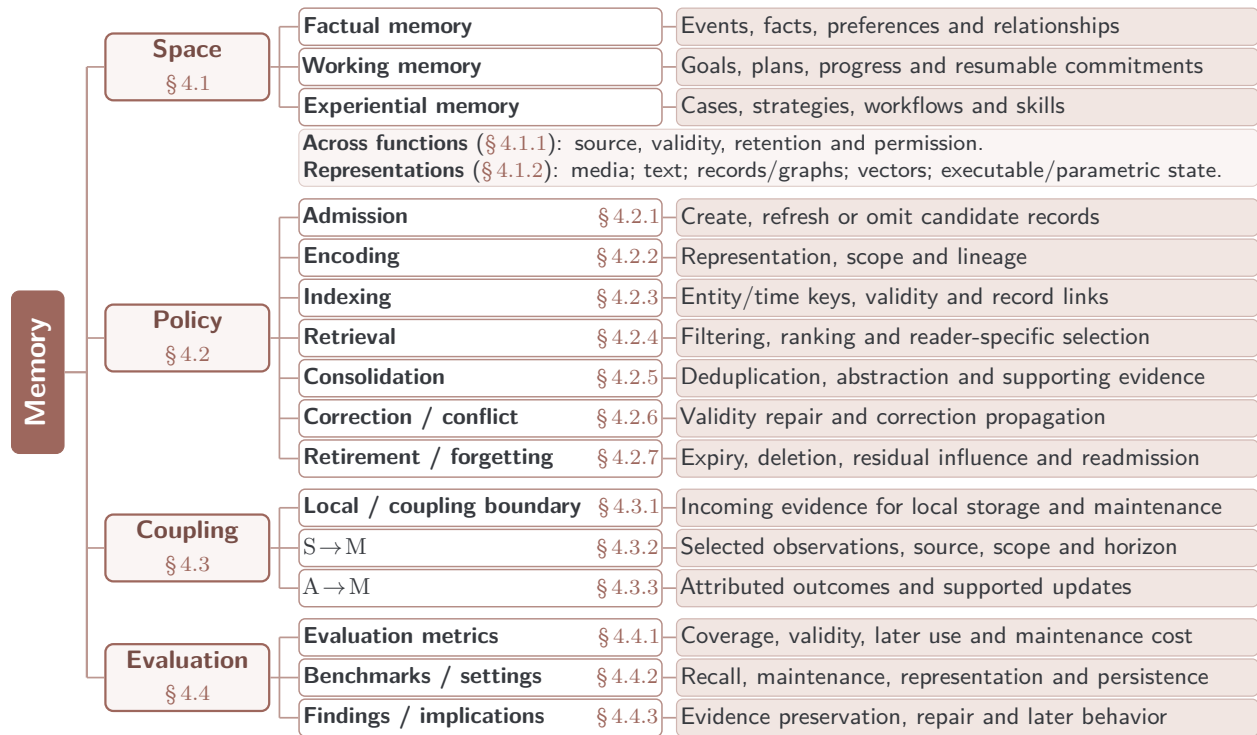


Figure 7: **Memory section structure and taxonomy.** Factual, working, and experiential memory, shared representations, local maintenance policies, incoming $S \rightarrow M$ and $A \rightarrow M$ evidence, and evaluation.

4.1 Memory Space

A preference and an unfinished task can outlive the same conversation but require different maintenance. Completing “send the report today” retires a commitment, while “avoid routine reminders during meetings” can remain relevant. Table 8 groups such information by its function in later decisions. Domain shapes its

media and storage locations, participant scope determines whose information and permissions it carries, and outcome horizon affects how long commitments and supporting evidence remain useful.

Long context expands the history available within a model call, with effective use varying by position and task [260, 261]. Context engineering selects the material supplied for the current task [262]. Across interactions, however, retained evidence may need to support different decisions as the user’s circumstances change. Persistent storage therefore involves choosing what to preserve, what to abstract, and what must remain correctable. The value of a retained record thus depends on the decisions it can support and the maintenance needed to keep it useful.

4.1.1 What Can Persist: Memory Functions

Following the functional taxonomy of agent memory [35], we distinguish factual, working, and experiential memory. These functions support knowledge about the user and environment, ongoing task state, and reusable experience. Table 8 connects their contents to continuing assistance. Persistence covers state for resuming tasks and knowledge or experience useful beyond their completion.

Table 8: **Memory functions and their roles in continuing assistance.**

Memory function	Typical contents	Role in continuing assistance
Factual memory	Events, facts, preferences, relationships	Keep personal context current
Working memory	Goals, plans, progress, commitments	Resume tasks across interactions
Experiential memory	Cases, strategies, workflows, skills	Reuse and revise past experience

Source, validity, retention, and permission apply across these functions. The working-memory row covers task state retained for later resumption.

Factual memory: events and knowledge. Episodic records preserve experienced events, while semantic records express facts abstracted from their sources [263]. MemoryBank [16] and Generative Agents [264] retain events and derive higher-level descriptions. Zep [265] and A-MEM [99] expose temporal or relational links, while PersonaVLM [266] extends personal context to visual observations. For continuing assistance, a past event and a current preference need different treatment: the event may remain relevant as history after the preference has changed. Source links allow later evidence to revise an interpretation while preserving what was originally observed.

Working memory: active and resumable task state. Working memory maintains task-relevant information under limited capacity [267]. The agent-memory taxonomy emphasizes active context management within an episode [35]. MemGPT/Letta [17] manages bounded model context with external storage, while MemoryOS [268] and MemOS [269] develop hierarchical memory infrastructures. An active workspace can be temporary. In continuing assistance, goals, progress, and unresolved commitments needed after an interruption or interaction boundary must survive in m_t so the workspace can be restored. Completing or cancelling a task can retire its active commitment while leaving an outcome record available for later interpretation.

Experiential memory: reusable cases and procedures. Experiential memory includes past execution cases, distilled strategies, and reusable skills [35]. Agent Workflow Memory [270] and Memp [271] make workflows and procedural records explicit. The grouping follows the use of experience: a failed reminder can be retained as factual history and also retrieved as a case when choosing a later strategy. Reuse depends on whether the earlier circumstances and procedure still apply. As routines, tools, or permissions change, past conditions and outcomes help distinguish reusable from obsolete guidance.

4.1.2 Representations and Storage

Representation and future questions. Across these memory functions, representation determines which details a future decision can recover. Raw media preserves evidence at high exposure and deletion cost; transcripts and summaries make content inspectable but may erase uncertainty; structured tuples and temporal graphs expose identity and validity assumptions; embeddings support efficient similarity search

but make corrections difficult to localize. Executable and parametric representations extend these choices to procedures and compact learned state. The following comparisons examine which representations preserve decision-relevant evidence and how their maintenance costs differ.

Choosing a storage architecture. A controlled comparison makes these representation choices empirical variables. Across dense and sparse indexes, textual, structural, and hierarchical stores, refinement-based memory, parametric updates, and activation-compatible context, Harness the Memory reports no uniformly dominant representation architecture: broad retrieval helps factual question answering, while excessive retrieval can hurt sequential decisions by displacing action-critical context [272]. For an assistant that both answers questions and executes tasks, how much to retrieve can differ by decision.

Compact learned memory raises a different retention question: which earlier facts survive successive updates? Metis [273] stores history in dense memory state while keeping its learned update mechanism fixed during interaction. Its synthetic-persona tests sequentially add personal facts and probe earlier information, finding declining recall as updates accumulate. The comparison connects compact storage to a continuing assistant’s need to retain earlier personal information while incorporating new evidence.

Personal content and decision rules. Consider “never remind me during meetings.” Retained as a user preference with provenance and validity, it is content in m_t . Retrieval supplies it to a fixed reminder controller. If an admitted change compiles it into a persistent rule governing future reminders or availability checks, that rule belongs to Σ_t . The operational role determines the classification, so an implementation can retain the preference and separately compile a control rule from it. The same distinction applies when personal evidence or decision rules are encoded in learned parameters.

Persistence and later use. Observation spends power, bandwidth, and exposure at capture time; persistence extends both the value and the maintenance cost of that evidence. An unretained event can become unrecoverable when its sensed representation expires, while a false admitted fact can survive its original context. Consolidation may spread that fact into a profile, summary, index, or procedure, increasing the work needed to correct or delete it. These delayed costs can grow despite inexpensive writes.

4.2 Memory Policy

Table 9 organizes the local Memory policy π_M around admission, encoding, indexing, retrieval, consolidation, correction, and retirement. We compare the available choices through what they preserve, their storage and read costs, and the reach of later repairs. These decisions operate on candidate content and retained state. They interact: encoding can remove details that retrieval later needs, and consolidation can change which records a correction must reach.

Table 9: **Local Memory decisions.**

Decision	Main choices	Condition and cost	Representative works
Admission	Create / refresh / omit	Uncertain future need: omission / repeated influence	AdaMem [274], PerMemBench [275]
Encoding	Episode / summary / typed record	Unknown future query: detail / read cost	Vinci2 [276], User as Code [277]
Indexing	Identity / time / links	Referent or validity changes: valid scope / stale links	Zep [265], A-MEM [99], RaMem [278]
Retrieval	Filter / rank / search further	Decision deadline: coverage / delay	GAM [279], RLMs [280], VoiceMem [281]
Consolidation	Summarize / derive / defer	Growing history: compact reads / lost lineage	LightMem [282], ReSum [283], AgentFold [284]
Correction	Amend / version / defer	Contradictory evidence: consistency / premature overwrite	MemConflict [285], STALE [286]
Retirement	Expire / redact / remove	Ended validity or access: deletion reach / later utility	Memora [287], MemPrivacy [288]

Admission determines whether candidate content creates or refreshes a record. Retrieval is a read. The condition/cost column identifies the decision pressure and trade-off. Works include methods and evaluations of that decision.

4.2.1 What to Store: Admission

Selecting for future relevance. Given candidate content, admission determines whether to create a record, refresh an existing one, or leave the store unchanged. AdaMem [274] learns personalized writing decisions, while Personalize-then-Store/PerMemBench [275] makes user-dependent storage relevance an evaluation target. Both address the fact that the same observation can matter differently to different people or tasks. Explicit user instructions, confirmation, and retention settings offer more direct control over what survives. A private utterance can support the current response without becoming a persistent record. The decision therefore affects future influence as well as storage volume.

Balancing omissions and repeated influence. Selective writing lowers storage and exposure, but rejecting a commitment can remove the only evidence that a later reminder needs. Broad writing preserves more possibilities while carrying uncertain identities or interpretations into subsequent decisions. The useful choice depends on the cost of those errors and on whether the original evidence remains available for reconsideration. Once content is selected for retention, encoding determines which details, uncertainty, and source links survive the write.

4.2.2 What Form Survives: Encoding, Scope, and Lineage

Preserved episodes and compact descriptions. Textual events and reflections support flexible recall [16, 264], while bounded-context management makes their read cost explicit [17]. Episode-preserving designs retain evidence for reinterpretation [289]. Summaries reduce the material needed for a later read, but select details before every future question is known. For continuing assistance, preserving the original episode is particularly useful when a later correction changes the interpretation of an earlier interaction. Its value must be weighed against the cost of storing, accessing, and eventually removing that additional content.

Source links and typed state. Vinci2 [276] links summaries to visual reconstruction, allowing queries to revisit compressed detail. User as Code [277] instead exposes typed fields for exact queries and alert conditions. The former supports questions that require revisiting evidence, while the latter makes known task conditions easier to evaluate. ABot-AgentOS [290] combines provenance-linked graph writes with current and historical state, evaluating failure-derived assets on later splits. These choices connect compact access to a maintenance question: when a preference changes, which sources and derived records can identify the state that needs repair?

4.2.3 How State Remains Reachable: Indexing and Identity

Finding the right person and time. Zep [265] represents temporal structure, A-MEM [99] links evolving notes, and Infini Memory [291] maintains topic documents. MOSAIC [292] combines entity organization with save-time conflict handling. These designs organize relevance beyond similarity: a statement may describe an earlier preference, a temporary exception, or another person. RaMem [278] explicitly anchors fragments to event time, mention time, session span, and participants, with validity-aware retrieval improving F1 across several backbones. Retaining these distinctions lets a later decision retrieve what applies now without erasing what was true before.

Maintaining links as evidence changes. A false merge can attach a valid preference to the wrong person, while duplicate records can make repeated evidence appear independent. Incremental maintenance saves rebuilding work but can leave stale links. CodeNib [293] compares repaired repository views with independently rebuilt targets and reports speedups for matching transitions. Although the indexed objects differ from personal profiles, the comparison supplies a concrete way to relate maintenance savings to consistency. For a continuing assistant, the corresponding concern is whether a newly corrected identity or task state becomes reachable through the same searches that previously found the old one.

4.2.4 What Is Exposed Now: Retrieval and Read Policy

Searching history within a response budget. GAM [279] uses concise session memos to navigate a full-record page store. Recursive Language Models [280] let code inspect external context and launch recursive calls during a query. Both separate accessible evidence from the immediate model context, trading additional search and computation for selective access. VoiceMem [281] starts retrieval from partial speech and defers stable-state consolidation until after the session. The costs fall at different times: further search can recover missed detail; anticipatory retrieval prepares known context before a response.

Selecting evidence for a particular consumer. A larger candidate pool can improve coverage while mixing evidence of different reliability. PGraphRAG [294] supplements sparse user histories with neighboring profiles, making an inferred preference different from the user’s own statement. LRAT [295] learns relevance from browsing choices and subsequent reasoning. DynamicMem [296] identifies retrieval as a major error source in long, multi-application histories. Together these studies make relevance depend on source, validity, and the decision being supported. A record useful for private ranking may be inappropriate for spoken output or for a tool that exposes it to another audience.

Using less personal context when appropriate. SteeM [297] makes reliance on memory user-controllable. OP-Bench [298] measures irrelevant personalization, repetition, and sycophancy, while The Compliance Trap [299] traces the propagation of conflicting memories. Additional personal context can therefore distract from a request as well as inform it. A selective read can omit a correct record when it is irrelevant to the present decision, too stale to trust, or unsuitable for its audience.

4.2.5 When Episodes Become Abstractions: Consolidation

Abstraction with access to its evidence. Consolidation turns repeated episodes into summaries, preferences, or procedures. Provenance-linked and buffered updates [300, 301, 291, 302] keep supporting evidence accessible as records change. Controlled update studies [303] and LifeDialBench [304] find settings where rewritten summaries are less useful than preserved episodes. A consolidated reminder preference can save repeated interpretation, but individual dismissals may still be needed to distinguish a recurring objection from one busy afternoon. The storage saving is useful when the abstraction preserves the distinctions later decisions depend on.

Scheduling maintenance outside the immediate response. LightMem [282] groups topics and defers maintenance, and sleep-time processing [305, 306] moves work away from the interactive path. ReSum [283] summarizes near a context limit, while AgentFold [284] selects historical steps and folding granularity. Mobile-Agent-E [307] derives Tips and executable Shortcuts from completed tasks. These approaches choose different

times and units for consolidation. Deferred work reduces immediate delay, while task-derived procedures can accelerate later execution. A continuing service must still use an available record during maintenance and carry subsequent corrections into any resulting abstraction.

4.2.6 Maintaining Current Validity: Correction and Conflict

Updating a value or preserving alternatives. A changed address or routine can supersede an earlier value without making the earlier report false. Versioning preserves that history, whereas a confirmed factual error can be amended. MemConflict [285] distinguishes temporal, factual, and contextual conflicts, and STALE [286] evaluates recognition of outdated memory. Keeping competing records unresolved is useful when source or scope remains unclear. For a pending action, a focused question can then be more appropriate than silently choosing the newest statement.

Propagating a correction. Updating the visible field repairs only the records reached by that operation. Stale-dependency evaluation [308] shows that summaries, cached reads, and later responses may retain an implicit dependency on the old value. Versioned records explain what was valid when, while descendant repair changes what later consumers receive. In continuing assistance, this distinction separates remembering that the user corrected a preference from actually following the corrected preference in the next interaction.

4.2.7 What Must Cease to Influence: Retirement, Forgetting, and Deletion

Ending validity and ending access. Expiration ends a retention interval, supersession deactivates an old assertion, and redaction removes selected fields. Deletion removes an object under the chosen storage semantics, while a minimal tombstone can prevent re-importing a revoked item. These choices serve different needs: task completion can retire a pending commitment while keeping its outcome, whereas a deletion request can require removing the underlying content. Reusable procedures can also lose validity through subsequent failures. In a stream of browser tasks, PANDO [309] distils reusable rules and parameterized routines after each task, demotes repeatedly failing skills, and records them in a persistent blacklist that blocks matching candidates from re-entering the library. Retirement governs continued use and readmission.

Removing residual influence. Memora [287] evaluates whether invalidated memories continue to affect subsequent answers and recommendations. Long-term-memory security work connects residual information to provenance and forgetting across the lifecycle [39]. A deleted source can survive in a summary, index, cache, or derived procedure, so source links help locate descendants that need removal or recomputation. If the evidence also supported a policy revision, removing content from m_t and rolling back the resulting change to θ_t are separate operations. Behavioral deletion concerns whether later decisions still depend on the invalidated information, beyond its disappearance from one store.

Learning the maintenance policy. MemSkill [310] separates trace-specific records from reusable memory-operation skills. A trained controller selects skills, a fixed executor applies them, and a designer revises the skill bank from difficult training cases, with rollback to stronger snapshots. This locates a local-policy revision in the reusable rules governing memory operations. Accumulating another record changes the content available to π_M ; revising an operation skill changes how subsequent records are handled, supplying a mechanism relevant to self-evolution.

4.3 Memory Coupling Across the Loop

Memory receives observations selected for retention through $S \rightarrow M$ and decision-linked outcome evidence through $A \rightarrow M$. The first dependency determines which observed details reach persistent state. The second determines what later results justify changing. We compare their inputs, the updates they support, and the difficulties of maintaining useful state across interactions. Table 11 summarizes representative mechanisms and their reported comparisons.

4.3.1 Local Decisions and Coupling Boundaries

Table 10 distinguishes incoming guidance from local Memory decisions. The local policy π_M implements admission, encoding, retrieval, and maintenance within the store. For $S \rightarrow M$, κ_{SM} selects sensed content and requests retention with its source, scope, and horizon; the receiving policy decides whether and how to create or refresh a record. For $A \rightarrow M$, attribution relates an outcome to an earlier decision and identifies a supported update. These rules can support content correction while remaining fixed. They are also possible self-evolving revision targets: changing the selection rule for future retention revises κ_{SM} , while changing how later outcomes are linked to earlier decisions and converted into attributed feedback revises κ_{AM} . Proposed policy changes pass through validation and durable commitment to θ_t .

Incoming observations and decision-linked feedback can inform the same update or be connected over time. A user’s completion report can supply both the task fact to retain and the evidence for retiring its commitment. It can also first be stored as an observation through $S \rightarrow M$ and later support the decision-linked update through $A \rightarrow M$.

Table 10: **Local Memory decisions and incoming guidance.**

Relation	Description	Example
Local policy	Selects and maintains records within Memory	Merge duplicate notes, preserving sources
$S \rightarrow M$	Sensed content is proposed for retention	A spoken commitment becomes a task record
$A \rightarrow M$	Attributed outcomes guide persistent updates	A user correction revises a preference

4.3.2 Sensing to Memory: the $S \rightarrow M$ Channel

Selecting observations for later use. Selecting an observation for retention determines which details Memory can preserve for later decisions. A spoken commitment needs its speaker, intended action, and relevant timing or completion conditions. A suggestion, an accepted commitment, and a report of completion can mention the same task but call for different updates. Preserving that distinction lets Memory decide whether to create, refresh, or retire a task record, even when the original conversation is no longer available.

Preserving source and permission. The transferred information also determines what Memory can later correct or remove. A summary detached from its source makes it harder to revise a claim when new evidence arrives. Disclosure controls operate on this transfer and the resulting state: Recall documents consent, encryption, and gated access [311], while MemPrivacy [288] replaces sensitive spans with typed placeholders and restores local values on device. The latter reduces what crosses a processing boundary, while storage controls restrict retained access. Missed sensitive spans and derived information remain distinct concerns. Provenance and descendant tracking help propagate corrections and deletion through the records built from those inputs [39].

Why observation-to-memory transfer is difficult. Future relevance is uncertain when an observation reaches the store. Omitting a commitment leaves no record for a later retrieval policy to find, while admitting a mistaken inference can carry it into repeated decisions. The repair depends on where information was lost. New evidence can complete an incomplete record, but more selective admission leaves an existing summary’s missing source link unresolved. AMemGym [312] separates writing, reading, and utilization, and WorldMemArena [313] combines multimodal memory points, updates, and distractors. These settings help distinguish failures to retain useful evidence from failures to use what was retained.

4.3.3 Action to Memory: the $A \rightarrow M$ Channel

Matching feedback to the state it can revise. Explicit corrections can identify a preference or task record that needs revision. ProPerAssistant [314] learns personalized recommendation policies from simulated ratings, while Reflective Memory [315], MMDuet2 [316], and Memory-R1 [317] connect outcomes to stored experience or memory operations. These mechanisms differ in the state they revise and how specifically the available feedback identifies what should change.

Selecting the object and timing of a revision. Feedback can change records, reusable procedures, or the policies that manage them [318, 319, 320, 321]. EmbodiSkill [322] distinguishes faulty skill instructions from failures to follow valid guidance: during evolution on training tasks, it revises defective instructions but records execution lapses in a supporting appendix while leaving valid rules in place. This avoids treating every failed action as evidence that the stored procedure is wrong. OPD-Evolver [323] and TMEM [324] distinguish explicit memory from policy updates, EvolveMem [325] revises retrieval configuration, and MemQ [302] propagates delayed utility over provenance links. These choices locate the lasting change: personal content in m_t or a decision rule in θ_t . A corrected task status and a revised rule for selecting future reminders consequently require different justification.

The point at which a change becomes active also matters. OpenClaw-RL [326] derives evaluative and corrective signals from subsequent interaction and publishes weights at controlled boundaries in simulated personal-preference experiments. RLinf-USER [327] combines replay with asynchronous real-robot training and automatic or human-assisted resets. Such designs separate collecting feedback from activating an updated policy. A continuing personal service needs that separation when earlier commitments remain pending while a new rule is being validated.

Why update influence is different from justified credit. Memory-R2 [328] re-rolls operations from matched intermediate memory states. HiMPO [26] compares recoverable task information before and after a write to limit credit leakage from other failures. These controls localize a memory operation’s contribution more closely than a shared task reward broadcast to retrieved candidates, as in Skill1 [329]. Other methods improve the feedback itself: AgentFlow [330] distributes terminal reward across planner turns, Web-Shepherd [331] learns checklist-supported step scores, and ZeroGUI [332] uses screenshots and voting to reduce false success rewards. ALTER in N_0 -VTLA [216] uses contact stages, drops, and human corrections on a fixed training corpus. More localized feedback helps identify an operation, while attributing a personal outcome also requires considering other plausible causes.

Freezing an update tests its downstream influence without validating the original credit assignment. A policy revision needs evidence that distinguishes its proposed explanation from competing causes; a directly identified record correction has a narrower target. Section 7.2 turns this distinction into an analytical reminder example. Delayed feedback and intervening causes motivate the qualitative High rating for $A \rightarrow M$ in Table 2.

Table 11: **Representative implementations of $S \rightarrow M$ and $A \rightarrow M$.** Rows connect incoming evidence to the persistent update it supports.

Work / protocol	Supplied information	Update supported	Reported evaluation
$S \rightarrow M$: observations selected for retention			
AdaMem [274]	Personal relevance	Write selection	Personalized writing and use
PerMemBench [275]	User-specific context	Storage admission	Personalized storage decisions
Imprint [171]	Personal event records	Record construction	Compression and later recall
Infini Memory [291]	Sources and topic state	Document updates	Maintenance and subsequent use
$A \rightarrow M$: decision-linked evidence supplied to updates			
Memory-R2 [328]	Re-rolled outcomes	Memory operations	Matched intermediate states
HiMPO [26]	Information gain	Write credit	Before/after-write comparison
EvolveMem [325]	Evaluation feedback	Retrieval configuration	Configuration revision
MemQ [302]	Delayed utility	Linked record updates	Provenance-based propagation
ImplicitRM [333]	Behavioral feedback	Reward estimation	Implicit-feedback modeling
OpenClaw-RL [326]	Next-interaction signals	Policy updates	Simulated preferences
RLinf-USER [327]	Robot experience	Replay and training	Real-robot task learning
AgentFlow [330]	Terminal reward	Planner updates	Turn-level reward assignment
ZeroGUI [332]	Screenshots and votes	Success rewards	GUI reward reliability

The receiving endpoint can be a record update, an update proposal, or the evidence used to evaluate one. Reward-estimation rows report that supporting step; the comparison column identifies what was tested.

4.4 Evaluation

4.4.1 Evaluation Metrics

Memory quality concerns coverage, current validity, and usefulness in later decisions. Extraction recall measures whether relevant evidence enters the store, while update accuracy and omission rates examine whether changes are reflected. Retrieval precision and downstream task results concern different endpoints: returning relevant content and successfully using it. Invalidation tests examine whether obsolete information stops influencing behavior. Measuring these outcomes alongside write and read latency, storage growth, and correction effort makes representation and maintenance costs visible. For continuing service, the order of writes, contradictions, and queries determines which state was available to each decision.

4.4.2 Benchmarks and Evaluation Settings

From recall to maintenance. Table 12 groups evaluations by the state and decisions they examine. LoCoMo [334] and LongMemEval [335] make conversational recall, temporal reasoning, and updates comparable. Personal-context suites broaden the histories and tasks [336, 337, 338, 339, 340, 341]. Operation-level tests ask what went wrong before an answer was produced. MemOps [342] supplies labeled transitions, and MINTEval [343] tests revision under multi-target interference. These protocols connect a final error to a particular maintenance operation.

Coverage before correction. Updates depend on earlier extraction; HaluMem [344] makes that dependency measurable. For the tested Mem0 configuration, extraction recall falls from 42.91% on HaluMem-Medium to 3.23% on HaluMem-Long; update omissions rise from 74.02% to 98.51%, although update hallucinations fall from 0.45% to 0.03% (source Table 3). The low hallucination rate accompanies very few successful updates, not reliable maintenance. Long adds irrelevant dialogue to Medium, so the contrast combines greater length and interference under the tested configuration. These results motivate tracking whether an update’s prerequisite fact was retained, alongside whether the update was correct.

Comparing what a representation preserves. MemLens [345] and MemEye [346] compare visual input with text, captions, or oracle evidence to expose information lost by a representation. Agent-native assessment [347] compares evidence access with its operation and latency costs. BrowseComp-Plus [348] fixes the document corpus and supporting evidence, while ExecRetrieval [349] distinguishes test-passing procedures from near-identical faulty ones. For memory used in action, usable evidence or an executable procedure is a more specific endpoint than semantic similarity.

LongMemEval-V2 [350] supplies a concrete cost comparison: removing AgentRunbook-R’s raw-state slice pool lowers Small-tier answer accuracy from 58.6% to 42.3%, while query latency falls from 26.9 to 16.7 seconds (source Table 2, Qwen3.5-9B controller and reader). The intervention removes an evidence pool and its retrieval work together. It therefore quantifies the value and query cost of this access path, without establishing a universally superior representation.

Testing later use after change. PAST-Bench [25] uses fresh-session tasks and persistence controls, while PATH-Bench [351] and CL-Bench [352] vary preceding histories. Memory and skill reuse are also tested through later executable tasks [353, 354, 355]. LifelongAgentBench [356], VitaBench 2.0 [357], and synthetic longitudinal HealthClaw tests [358] make changing personal state part of the workload. RNG-Bench [359] compares remembered cards or visited regions with externally supplied state. Its controls can reduce perception and representation demands as well as recall, showing why the exact information supplied by a baseline matters. Prospective-memory tests [360, 361] further ask whether a commitment resurfaces when its condition is met. These comparisons show when prior experience helps or hinders later decisions.

Table 12: Memory benchmarks and evaluation settings.

Benchmark / protocol	Evaluation setting	Evaluated content
Memory space: evidence and representations		
LoCoMo [334]	Multi-session conversations	Long-term conversational recall
LongMemEval [335]	Multi-session histories	Temporal reasoning, updates, abstention
MemLens [345]	Multimodal memory tasks	Image-removal and oracle controls
MemEye [346]	Visual-state changes	Text, caption, and oracle-image controls
LifeDialBench [304]	Online lifelog memory	Episodes and summary updates
LongMemEval-V2 [350]	Long agent histories	Evidence access and latency
Local policy: maintenance and suitable retrieval		
PerMemBench [275]	Personalized storage	Future-relevant content selection
MemOps [342]	Labeled memory transitions	Trigger, target, and update accuracy
HaluMem [344]	Extraction, updates, and QA	Hallucinations and omissions
MemConflict [285]	Conflicting memory records	Conflict handling
STALE [286]	Outdated personal records	Stale-memory recognition
Memora [287]	Invalidated memory	Invalidation-aware forgetting
MemPrivacy-Bench [288]	Sensitive content in memory	Masking and retained memory utility
AMemGym [312]	Interactive memory operations	Writing, reading, utilization
MINTEval [343]	Multi-target revisions	Interference and updates
ExecRetrieval [349]	Near-identical procedures	Test-based retrieval utility
Coupling and later use: state influences subsequent decisions		
MemoryArena [24]	Interdependent subtasks	Memory construction and action use
StreamMemBench [362]	Streaming input and feedback	Future-oriented assistance
PAST-Bench [25]	Cross-session persistence pairs	Later task benefit and interference
PATH-Bench [351]	Helpful/interfering histories	Transfer and forgetting
CL-Bench [352]	Stateful task sequences	Learning gain over prior capability
EvoMemBench [353]	Within- and cross-episode tasks	Knowledge versus procedural memory
BenchTrace [354]	Annotated failure episodes	Reflection quality and failure avoidance
ContinualSkillBench [355]	Ordered, interdependent tasks	Skill reuse versus context-only learning
MemoryAgentBench [336]	Incremental agent memory	Retrieval, learning, forgetting
WorldMemArena [313]	Multimodal memory updates	Use under changes and distractors
RNG-Bench [359]	Remembering-to-act tasks	External-state and history controls
TriggerBench [360]	Latent commitments	Timely recall without false alarms
PM-Bench [361]	Simulated weekly activities	Prospective commitments

4.4.3 Findings and Implications

The comparisons favor different representations for different future uses. Preserved episodes support reinterpretation, while summaries and typed records reduce the cost of selected decisions. HaluMem’s operation-level results show why accurate retrieval cannot compensate for an unwritten fact. MemPrivacy-Bench’s masking-utility comparisons [288] show a cost of restricting access: removing sensitive information can also remove evidence needed by a later answer.

Longitudinal studies locate problems at different origins: repeated updates can erode retained capability [363], summarization can hide contamination [364], and routine interaction can corrupt admission [365]. A committed claim can sustain sycophantic behavior [366], while a visible correction can leave an implicit dependency intact [308]. These findings support different repairs. Stricter admission cannot fix an orphaned summary, and better retrieval cannot restore provenance lost during encoding. Following corrected state into later behavior is the comparison that reveals whether the repair reached its intended use.

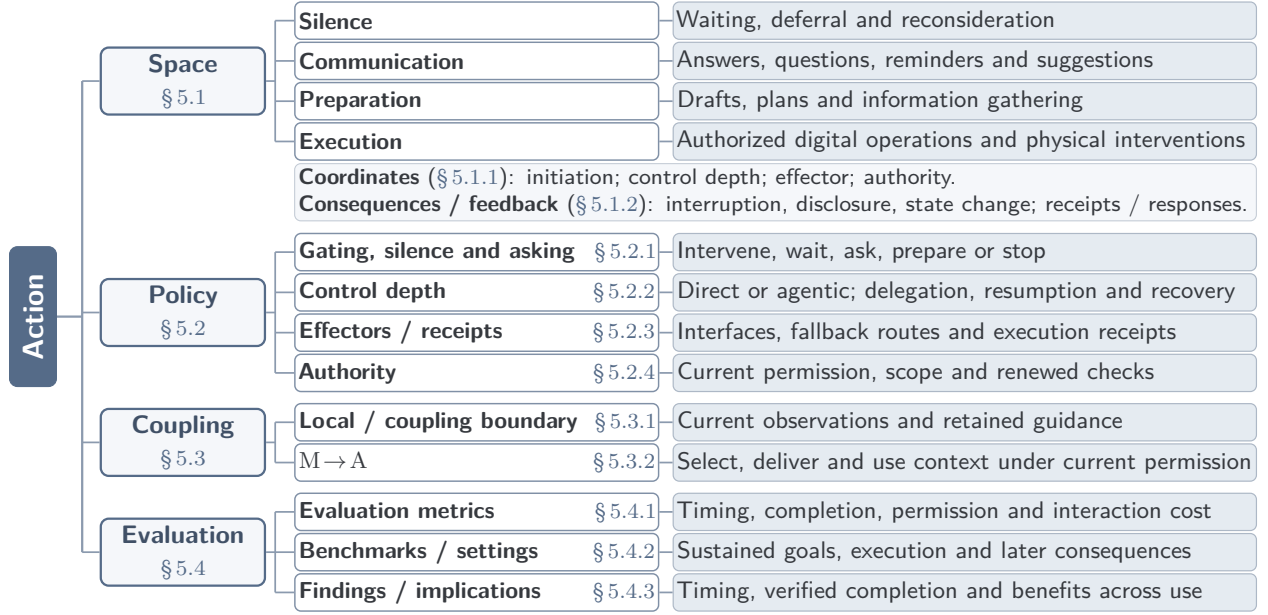


Figure 8: **Action section structure and taxonomy.** Interventions, action coordinates and consequences, local policies, incoming $M \rightarrow A$ guidance, and evaluation.

Memory Must Remain Usable and Correctable. Memory supports continuing assistance by keeping preferences, evidence, and unfinished commitments usable and correctable. Admission occurs under uncertainty about future relevance: an omitted commitment leaves no record to retrieve, while an admitted mistake can recur in later decisions. Evaluation should follow admitted facts, later corrections, and invalidated records and their descendants through to their effects on subsequent assistance.

5 Action

Action supports a continuing loop by considering assistance between explicit requests and resuming unfinished commitments from durable task state. Waiting, preparing, continuing, and cancelling remain decisions even when no new request arrives. Figure 8 organizes the section around available interventions, local policies, coupling, and evaluation. We compare when assistance is worth initiating, how control remains within current permission, and how interruptions and recovery costs accumulate across interactions. The self-evolving extension asks when outcome evidence justifies revising the intervention rules or the way retained guidance is selected for future actions.

5.1 Action Space

For a reminder assistant, notifying the user, preparing a document, and sending it are different interventions. They require different evidence and permissions and create different commitments. AIMA [88] frames action selection around expected performance, while Horvitz’s mixed-initiative framework [367] includes the cost of interrupting or overriding the user. In continuing service, the comparison includes the value of waiting and the attention, access, and recovery costs that accumulate across interactions.

5.1.1 What Can Be Done: Interventions and Interfaces

The action space includes silence, communication, preparation, and execution. Asking, answering, reminding, and recommending are communication examples; drafting is preparation, and delegated tool use is execution. ProAct [238] illustrates preparation between requests by acquiring information for likely upcoming needs. Recovery or escalation can combine these behaviors. Their effects may concern attention, information

disclosure, digital resources, or the physical setting. A private notification and a shared-document edit can serve the same goal while involving different audiences and permission requirements.

Table 13 describes an action through its initiation, control depth, effector, and authority. Current permission bounds the scope of the selected intervention. Initiation and control depth describe different properties: a user-requested GUI workflow is reactive and agentic, while an unsolicited one-sentence warning is proactive and direct. Under this initiation coordinate, following a standing instruction remains reactive: its goal or watch condition comes from the user, even when execution occurs between requests.

Table 13: **Four coordinates of an action.** Initiation records the opportunity source; the other coordinates describe control, interface, and permission.

Coordinate	Example values	Describes	Example
Initiation	Reactive / proactive	Opportunity source	Request / detected need
Control depth	Direct / agentic	Control across steps	Reply / delegated task
Effector	Dialogue / GUI / API / physical	Execution interface	Message / tool operation
Authority	Purpose-specific grants	Permission scope	Draft / approved send

Interfaces and observable effects. Dialogue, notifications, AR overlays, and haptics address attention or knowledge. GUI and API operations change digital resources, while physical effectors change location, viewpoint, or objects. VisionClaw [10] illustrates situated digital execution: observations through smart glasses can support creating notes from physical documents or calendar events from posters. The available interface determines whether the assistant can offer information, prepare material, or commit a change to another service. These options differ in the permission needed, the feedback returned, and the cost of recovery if the user changes plans or execution is interrupted.

5.1.2 What Follows an Action: Consequences and Feedback

An intervention can interrupt the user, disclose information, or change external state. A private draft may be easy to undo, while a public question can expose sensitive context and a sent message may already have been read. Silence also has consequences: it can avoid interruption or leave an opportunity unmet. These effects can become visible after the interaction ends. Observed change alone does not identify its cause or value: the user may complete a task independently while the assistant remains silent. Subsequent observations and directly delivered feedback provide the evidence for deciding what to do next.

Feedback from execution and interaction. API receipts, errors, file differences, and test results reveal different aspects of execution. Return-interface studies show that the detail supplied affects subsequent performance [368, 369]. A success receipt can confirm acceptance while leaving the final effect to be checked. User responses address whether the intervention was welcome or useful. Alongside explicit corrections, behavioral signals can supply feedback: ImplicitRM [333] uses abandonment, edits, and partial copying. A correction can identify a problem; behavioral rejection usually leaves more uncertainty. Feedback therefore helps Action choose whether to continue, clarify, retry, or wait. For feedback arriving after an interaction, retained decision and execution context supports $A \rightarrow M$ attribution and persistent updates.

5.2 Action Policy

Table 14 summarizes the local choices governing whether, when, and how to intervene. We compare their information needs, interruption and execution costs, and effects on completing or resuming a task. The opportunity’s origin is given by the request or trigger. Using current state, retained guidance, anticipated consequences, and shared conditions, π_A selects a response and resolves its timing, control, interface, and permitted scope. These choices interact: a relevant response can arrive too late, and a successful operation can still exceed current permission.

Table 14: **Local Action decisions and representative works.**

Policy	Main choices	Condition and cost	Representative works
Gating / timing	Wait / clarify / prepare / proceed	Uncertain receptivity: missed opportunity / interruption	ProActor [370], ProAct [238], HeartSteps [371]
Control depth	Direct / plan / delegate / resume	Unfinished work: recovery / coordination overhead	ReAct [19], ReCode [372], AOrchestra [373]
Effector	Dialogue / GUI / API / physical	Available interface: operation cost / quality of receipts	VisionClaw [10], ToolOrchestra [374]
Authority	Proceed / confirm / withhold	Current grant: confirmation effort / unauthorized effect	AGrail [375], Formal enforcement [376]

Initiation records the opportunity source in Table 13. The local policy chooses what to do about it.

5.2.1 Whether and When to Intervene: Gating, Silence, and Asking

Using an expressed or inferred goal. A request or standing instruction supplies an expressed goal, while a detected opportunity first requires deciding whether the inferred need is relevant. Dialogue initiatives and help offers [40, 42] study this additional choice. For “remind me after the call,” the goal is given, but the deadline and current task status still affect whether to notify, clarify, or close the commitment. An inferred need can instead favor checking receptivity or offering help before execution. Both cases require comparing the expected value of intervening with waiting.

Waiting for a useful moment. Interruption-aware work compares assistance with waiting for a less disruptive opportunity [377, 378, 75]. Speech and video studies make silence, activation windows, and turn-taking explicit targets [379, 380, 258, 381, 382]. ProAIDE’s five-day IDE study [13] connects receptivity to the current workflow. For a reminder, waiting preserves an opportunity if a meeting ends before the deadline; repeated deferral can make the same content irrelevant. Recent interruptions and unresolved commitments therefore matter alongside the quality of the next response.

Clarifying, preparing, or stopping. Deferral serves different purposes. Robix [383], RoboOmni [384], and AmbiK [385] examine clarification before commitment. Agentic Abstention [386] and Gnosis [387] use stopping or confidence signals during execution. A question can resolve an ambiguous goal, preparation can make a future response cheaper, and stopping can prevent an uncertain action from continuing. The appropriate choice depends on whether the obstacle is missing information, a poor moment, or an unsafe next step.

Separate and shared timing models. Proactive Agent [12] learns welcome offers from simulated event streams, while Communication Policy Evolution [388] revises communication choices through simulated-persona rollouts. These studies examine learning or revising intervention rules. At runtime, ProActor [370], trigger-and-routing models [389], LlamaPIE [390], and PRPF [180] select opportunities before expensive generation. StreamBridge [100] places activation in a separate model, whereas MMDuet [391] adds informative and relevance heads alongside its language head to decide when to start a response. A separate activation model can be revised without changing the response model; shared representations tie timing decisions to the same video-language state used for generation. Runtime coordination trades computation during waiting against response delay and missed opportunities, while self-evolution additionally concerns whether a revised rule improves subsequent intervention decisions.

5.2.2 How Control Is Retained: Direct and Agentic

Retaining control across steps. A direct policy resolves one bounded decision; an agentic policy retains control across a plan, observes intermediate effects, and recovers when needed. ReAct [19] supplies the interleaving pattern. More retained control creates opportunities for recovery at additional inference cost. Efficient Agents [392] varies the ReAct step limit with replanning at every step: GAIA accuracy is 41.82%, 52.73%, and 53.33% at limits of 4, 8, and 12, with reported token use of 88K, 171K, and 243K (source Table 3).

The last budget increase yields little aggregate gain. These bounded-task results motivate weighing control depth against marginal cost; a continuing service also budgets for background work and other commitments.

Delegation and bounded context. Control can be distributed in different ways. LongHorizon-Harness [393] keeps task state in a manager while fresh-context executors act and a read-only auditor checks results. ReCode [372] expands plan functions during execution, changing the granularity of control. AOrchestra [373] selects each delegate’s instructions, context, tools, and model. These designs trade coordination overhead against bounded context and specialized execution. Evidence must stay current across handoffs: EnvTrustBench [144] shows how stale or adversarial environment content can mislead decisions.

Resuming an unfinished commitment. Across interactions, the control problem also includes resumption. A reminder or delegated task may remain pending while the assistant sleeps, a tool is unavailable, or the user changes plans. Memory retains the commitment and completed steps; Action chooses whether to resume, revise, cancel, or seek confirmation. Resumption needs the current task state and permission. If a submission timed out, checking whether it already succeeded can prevent a duplicate attempt. If the user completed the task elsewhere, the appropriate next step may be to close the commitment. These cases connect local execution control to observation through $A \rightarrow S$ and to durable task maintenance through $A \rightarrow M$. A task can finish or be cancelled while the continuing service remains available for other commitments.

5.2.3 Through Which Interface the State Changes: Effectors and Receipts

Choosing an execution surface. Interface selection connects the intended effect to an available operation and its feedback. ToolOrchestra [374] learns tool and model choices against monetary cost, latency, and tool preferences. SkillOrchestra [394] instead consults a handbook of competence and execution costs learned from traces. These mechanisms select among routes whose execution costs and available evidence differ, particularly when a device or service becomes unavailable.

Selecting evidence for the next step. Execution cost includes resolving uncertainty left by the interface. A low-cost operation with an uninformative return can require another observation or a user check before work can continue. The useful choice therefore includes the evidence available for continuation and recovery. For a continuing assistant, an unresolved result can require a later check before deciding whether to retry.

5.2.4 Which Commitment Is Authorized: Authority

Permission at the commitment point. Authority bounds an action by resource, audience, time, and permitted commitment. A grant may permit private advice or drafting while reserving a send or file change for confirmation. When evidence is weak, the policy can select a draft, clarification, or silence within current permission. A stored instruction can restrict behavior or record a grant, while c_t determines what is permitted now. Adaptive checks and explicit enforcement address this boundary differently. AGrail [375] generates and refines safety checks across tasks. Formal policy enforcement [376] checks declarative rules at relevant actions under an environment-observability contract. CRAFT and τ -break [395] test persuasion against tool-use policies. Restating the policy in the prompt reduces some violations while leaving enforcement failures. These results motivate checking actual permission at the commitment point.

Rechecking after a pause or handoff. Continuing tasks make the time of that check important. Permission to prepare a message can remain valid while permission to send it expires or is revoked during a pause. The next execution decision must use the current grant and audience, including after a tool handoff. A preview, renewed confirmation, or cancellation can then be selected according to the remaining uncertainty. Recovery depends on the effect: deleting a sent message may remove its record without reversing disclosure.

5.3 Action Coupling Across the Loop

The local Action policy decides whether and how to intervene using the current situation, retained guidance, and shared conditions. Here we focus on $M \rightarrow A$: which personal context reaches that decision, how it is

presented, and whether it remains appropriate when action is considered. Table 16 compares representative implementations and the receiving choices they evaluate.

5.3.1 Local Decisions and Coupling Boundaries

Table 15 distinguishes the receiving policy from its information sources. Current observations reach Action through the basic $S \rightarrow A$ path; retained personal state selected and delivered for the decision supplies $M \rightarrow A$. The local policy resolves the intervention using both under current conditions. A call-end cue can make a reminder timely, while a retained completion record removes the need to send it. Earlier channels can contribute to either input: a commitment may first be stored when observed, and personal context may help Sensing interpret the new scene. The boundary is the information supplied to this action decision, even when several channels participate in producing it.

Table 15: **Local Action decisions and incoming guidance.**

Relation	Description	Example
Local policy	Chooses the intervention and its timing	Defer while the user is occupied
$S \rightarrow A$ baseline	Current observations guide the action	A call-end cue makes a reminder timely
$M \rightarrow A$	Retained personal state guides intervention	A completion record suppresses a reminder

5.3.2 Memory to Action: the $M \rightarrow A$ Channel

Using personal and task context. Retained context can affect both whether to intervene and how to perform a task. A recent-reminder record can suppress redundant advice, while task history can guide resumed work. In either case, π_A combines the delivered guidance with current circumstances and permission. MementoGUI [396] supplies selected and compressed task history to GUI decisions and compares GUI task performance against memory baselines. MEMENTO [397] instead uses oracle perception and motor skills in simulated household tasks to focus on personal-memory use. Their settings expose different sources of uncertainty: GUI performance depends on the whole task pipeline, while the household configuration controls perception and execution to examine retained context.

Delivering reusable procedures. MMSkills [398] supplies procedures, state cards, and reference imagery against a screenshot, whereas LatentSkill [399] exposes selected procedures through cacheable adapters. This comparison concerns inspectable context and compact delivery. Both need the procedure’s preconditions to remain applicable when a paused task resumes or the application state changes.

Separating retrieval, delivery, and use. Local retrieval makes candidate records available. $M \rightarrow A$ selects and presents guidance to the Action decision, whose policy may consume it without changing the eventual choice. *Demystifying Agent Skills* [400] separates ranking, selection, and execution using matched source trajectories. This makes different repairs visible: a missing candidate calls for retrieval changes, an irrelevant delivered procedure calls for better selection, and a misapplied procedure calls for a change in its use. Substituting only delivered context tests the receiving decision more narrowly than replacing the store, which also changes retrieval and other consumers. An unchanged response can reflect sufficient current evidence or redundant records. A new preference can alter an intervention under the same delivery rule. A self-evolving revision to κ_{MA} instead changes how retained guidance is selected or presented across later actions; revising π_A changes how Action uses the supplied guidance to choose an intervention.

Why relevant memory can still misdirect action. An old quiet-morning preference can conflict with a nearing deadline, and valid facts can be irrelevant to the present task. PACE [401] evaluates retrieving personal facts and events to identify request conflicts. Studies of reliance, irrelevant personalization, and profile-memory interactions [297, 298, 402] further show why retrieval quality alone cannot determine whether to use a record. The difficulty is matching retained guidance to the current decision when relevance, user needs, and validity change at different times.

Authority adds a distinct condition on use. EAL-Bench [403] replaces memory while holding the request, executor, tools, and authoritative policy fixed, using exact repairs and sham edits as controls. Its synthetic organizational scenarios isolate memory-induced permission errors. A retained assertion can help select among permitted actions but cannot create a new grant. When current observations and grants are available, Action can recheck retained guidance before commitment. The qualitative Moderate rating in Table 2 reflects this matching problem, whose difficulty increases as records, needs, and permissions diverge.

Table 16: **Representative implementations of $M \rightarrow A$.** Rows connect retained guidance to its use in Action and the reported comparison.

Work	Supplied information	Use in Action	Reported evaluation
$M \rightarrow A$: personal and task state guide intervention			
User as Code [277]	Typed personal state	Select alerts	Alert-pipeline variants
PASK [109]	Long-term context	Proactive assistance	Short tasks and memory cases
MEMENTO [397]	Personal task memory	Household decisions	Oracle perception and skills
MementoGUI [396]	Selected task history	GUI decisions	Task and memory baselines
MMSkills [398]	Procedures and state cards	Ground a stored skill	GUI and game benchmarks
LatentSkill [399]	Selected skill adapters	Reuse procedures	Success and context cost
PACE [401]	Personal facts and events	Assess request conflicts	Retrieval/conflict accuracy
EAL-Bench [403]	Memory under fixed rules	Check permission	Memory swaps; fixed authority

5.4 Evaluation

5.4.1 Evaluation Metrics

Action evaluation connects the chosen intervention to its timing, consequences, and cost. Timing measures distinguish missed opportunities, premature responses, avoidable delay, and appropriate silence. Task outcomes separate an issued operation from completed work and user benefit. Permission checks assess the grant in force at each commitment point, while execution latency, computation, interruption, and recovery effort expose the cost. Across interactions, later completion, repeated prompts, correction effort, and changing receptivity reveal consequences beyond the immediate response [76, 77]. Privacy complaints and reports of social discomfort add evidence about harm, although their coverage depends on what users notice and report. Each evaluated consequence needs an observation horizon suited to the promised service.

5.4.2 Benchmarks and Evaluation Settings

Timing and continuity. Table 17 separates response timing, sustained goals, execution, and consequences. EgoServe [276] matches category-specific intervention windows before assessing grounded content, whereas ProReady-QA [246] scores responses against the availability of answer evidence. These protocols expose different timing errors. Longer-running settings ask whether the system continues tracking an earlier commitment: SentinelBench [259], IPIBench [241], and ARE/Gaia2 [408] introduce background events, changed instructions, and deadlines. Evaluations of unsolicited coding insights and silence [425] add the decision to leave the user undisturbed. The relevant comparison is between responding to an opportunity supplied now and continuing to act on a goal as its conditions change.

Attempts, completion, and permission. Digital and embodied task suites make execution measurable [20, 426, 427, 21, 140, 428], but their endpoints differ. MCPMark [414] checks resulting application states and repeated success. MCP-Universe [415] combines static grading with live-reference evaluators. MobilityBench [421] replays recorded API responses and checks arguments, excluding ambiguous requests and clarification; ClawBench [422] retains live sites but intercepts the annotated final submission. The latter can assess the proposed submission without demonstrating its final external effect. HarnessAudit [416] and Claw-Eval [417] use traces and policy checks to expose violations missed by completion scores.

Table 17: **Action benchmarks and evaluation settings.**

Benchmark / protocol	Evaluation setting	Evaluated content
Local policy: response timing and sustained goals		
EgoServe [276]	Egocentric intervention windows	Timing and grounded response quality
Pro ² Bench [230]	Procedural assistance streams	Guidance timing and recovery
ProVoice-Bench [404]	Voice interaction streams	Speak/silence timing
STRIDE [405]	Continuous video interaction	Response timing
AURA [406]	Reactive and proactive settings	Context and response quality
OmniPro [407]	Synchronized audio and video	Multimodal proactive responses
ProReady-QA [246]	Answer-evidence time windows	Early and delayed responses
SentinelBench [259]	Standing-request monitoring	Responses to background events
IPIBench [241]	Added, revised, cancelled requests	Tracking changing standing requests
ARE/Gaia2 [408]	Stateful apps, simulated time	Asynchronous work and deadlines
PersonalHomeBench [409]	Personalized smart homes	Context-appropriate assistance
ProAgentBench [410]	Real interaction events	Welcome intervention timing
PI-Bench [411]	Earlier personal interactions	Later proactive intent resolution
ATRBench [412]	Future-session information needs	Anticipatory information requests
PROBE [413]	Unprompted task opportunities	Task discovery and completion
TriggerBench [360]	Latent commitments	Timely action and false alarms
PM-Bench [361]	Week-long simulated activity	Prospective task completion
Execution and coupling: effects, context, and permission		
MCPMark [414]	Executable tool workflows	Final states and repeated success
MCP-Universe [415]	Live tool services	Static/live-reference grading
ST-WebAgentBench [147]	Permission-constrained web tasks	Authorized task completion
HarnessAudit-Bench [416]	Constrained execution traces	Permission and data-flow breaches
Claw-Eval [417]	Traces, audit logs, snapshots	Completion, safety, robustness
PRISK [402]	Personalization-sensitive tasks	Inappropriate personalization
PACE [401]	Hidden request conflicts	Conflict retrieval and judgment
EAL-Bench [403]	Memory swaps, fixed authority	Memory-induced authority violations
OpenComputer [418]	Application-state verification	Externally checked task outcomes
MiroEval [419]	Execution processes and reports	Report grounding and alignment
HumanCLAW [420]	Simulated physical skills	Completion under disturbances
MobilityBench [421]	Recorded API replay	Planning and argument correctness
ClawBench [422]	Live sites; final submit intercepted	Submission payload and attempt
Consequences and cost: outcomes beyond the issued action		
SoCRATES [423]	Simulated mediation pairs	Intervention timing and consensus
Claw-SWE-Bench [424]	Harness variants and resets	Execution under tested configuration
Efficient Agents [392]	Inference-budget variants	Task success per inference cost
HeartSteps MRT [371]	Randomized daily suggestions	30-minute activity response

Benefits and costs across repeated interactions. HeartSteps [371] reports a six-week micro-randomized trial with 44 adults, measuring steps in the 30 minutes after each randomized decision point. Walking suggestions versus no suggestion increased steps by 24% on average (59 steps; $p = .02$); the initial effect was 107% (271 steps; $p < .01$) and diminished over time. The changing proximal effect motivates repeated measurement, without identifying fatigue as its cause or establishing long-term health benefit. SoCRATES [423] instead compares mediated and unmediated simulated conflicts through later consensus. Claw-SWE-Bench [424] examines runtime configurations. These settings distinguish user outcomes from execution performance and identify the costs and temporal scope that must accompany a gain.

5.4.3 Findings and Implications

The comparisons support three practical distinctions. Response windows help choose a useful moment; execution checks distinguish an attempted operation from completion; repeated-outcome studies reveal benefits and costs that change with use. Together, they support choosing response timing, verification effort,

and recovery options according to the service’s expected consequences. Declining receptivity can favor fewer interruptions even when individual reminders remain relevant.

The remaining service-level question is whether a local gain preserves other commitments. Fewer interruptions can improve receptivity while allowing deadlines to pass; a longer execution plan can complete one task while delaying another. Evaluations should therefore follow pending, completed, and cancelled commitments across pauses, together with the checks and user effort needed to resolve them. If outcome evidence warrants a policy revision, its later comparison must include both the targeted improvement and commitments that the new rule might neglect.

Action Must Account for Accumulated Costs and Current Permission. Between requests, Action still chooses among silence, communication, preparation, and execution, weighing expected benefit against accumulated interruption, disclosure, and recovery costs. Unfinished commitments may need to be resumed, revised, or cancelled as circumstances change. Retained guidance cannot expand permission: current authorization must be rechecked at commitment, including after a pause or handoff.

6 Applications, System Bottlenecks, and Loop Closure

We follow Sensing, Memory, and Action decisions through service cases, comparing the evidence and operating conditions they require. Each application discussion connects reported mechanisms to system bottlenecks and possible repairs. Closure levels organize the requirements for continuing and self-evolving service.

6.1 Applications and Bottlenecks

Following Section 2.5, we compare cases through interaction domain, participant scope, and outcome horizon, $r = (b, n, h)$. Domain locates observation and action, participant scope identifies the people involved or affected, and horizon specifies when the evaluated consequence becomes visible. These coordinates expose different resource, permission, and feedback conditions. We group cases by service task into four families matching Table 18, which records the coordinates of each selected case. Within each family, we examine how those conditions shape the use of current observations, retained context, and action outcomes.

6.1.1 Situated Daily Assistance

Situated assistants connect egocentric observations to help during everyday activity. Procedural-assistance surveys [429] organize step understanding and error detection, while EgoLife [127] provides daily-life streams for studying longer personal histories. Vinci [8] and ProAssist [430] emphasize timely responses to current visual context. Sensible Agent [431] selects context-sensitive content and response modalities, and ProMemAssist [9] models transient working memory for Sensing-side state estimation.

The timing requirement differs between immediate guidance and retrospective recall. Within a procedural task, Pro²Assist [432] checks step predictions for temporal consistency before updating its progress record, and suppresses repeated guidance while the step and execution status remain unchanged. LightMem-Ego [433] separates current and short-term updates from asynchronous consolidation, using later transcripts to refine event records. Procedural guidance needs a stable current-step estimate, while retrospective recall can benefit from later refinement. This comparison identifies which evidence must be ready for the immediate decision and which maintenance can proceed afterward.

Bottlenecks and trade-offs. Resource studies measure sensing and inference costs [157, 155] and streaming processing efficiency [434]; interaction benchmarks examine response timing [435, 436]. Increasing capture can preserve more events while adding transmission, storage, and maintenance work. Reducing capture can shift the limitation from acquisition cost to missing personal history. A wrong association already retained requires a different repair: Memento’s multi-day study [220] reports useful contextual recall alongside mismatched visual referents, and its targeted removal control lets users stop unwanted memories from recurring. Thus observation coverage and record correction address different failures. Continuing daily assistance needs timely evidence and records that remain correctable across interactions.

6.1.2 Conversation and Meeting Assistance

Conversation combines concurrent input with potentially several participants. Full-duplex research supplies turn coordination [437, 438, 439], while LlamaPIE [390] and SocialMind [239] study discreet or context-sensitive assistance. A private cue and a spoken suggestion can carry the same content but differ in interruption and disclosure. The service consequently depends on who is captured, who receives the response, and whether retained context is appropriate for that audience.

Bottlenecks and trade-offs. A delayed background result may refer to a topic or audience that has changed, while an immediate response may interrupt a useful exchange. ProVoice-Bench [404] and long-term streaming voice tasks [440] provide timing and continuity tests. Increasing intervention frequency also spends attention and creates more outcomes to interpret. Across sessions, a meeting-derived commitment may remain relevant even though the people entitled to receive its details differ. Timely help therefore depends on both fresh conversational evidence and selective use of retained context. When the audience changes, restricting memory reads to records appropriate for the current participants and rechecking the intended recipients before delivery can prevent stale audience assumptions from guiding a response. When the topic changes, revising or withholding a queued response addresses its timing and relevance.

6.1.3 Digital Workspace and Personal Services

Digital services carry plans and personal context across applications and interactions. Some also retain revisions to the procedures or code governing future work. The digital-colleague perspective [441] connects persistent workspaces, reusable skills, and governance. PITCH [245] makes continuity concrete through scheduled morning planning and evening reflection in Slack: the evening exchange reuses the morning conversation to ask about progress and departures from the user’s plan. Its field-study dialogues include refusals and misplaced follow-ups. Retained plans can focus later help, but their suitability depends on the user’s current situation and receptivity.

Product descriptions extend this pattern to background work. ChatGPT Pulse [442] describes proactive personalized updates, and Claude Cowork [443] describes delegated workspace tasks. Muse [15] combines scheduled and event-driven work with memory across conversations and selective notification [444], with Sentinel checks for permission-sensitive operations [68]. Windows Recall [311] illustrates the related storage boundary: capture controls, protected access, and deletion determine which screen history remains available for later use.

Persistent workspaces also support reusable procedures. ReasoningBank [445] and Agentic Context Engineering [84] retain feedback-derived guidance, while Skill-Guided Continuation Distillation [446] and SkillHarness [447] develop reusable GUI procedures. KnowAct-GUIClaw [448] connects retrieval of experience and skills to structured subtask results, interface-state checks, and procedure repair from execution feedback. Its evaluation tests a joint host-router-memory increment and a separate skill increment, measuring the effects of these bundles. SE-GA [449], ScaleWebAgent [450], and Darwin Mobile Agent [451] provide related experience-reuse mechanisms. A retained procedure forms part of the persistent policy when it governs later execution choices. As records and procedures move between tasks, selecting context for the current audience and propagating corrections become part of maintaining a usable workspace.

Bottlenecks and trade-offs. Keeping that workspace available involves different resource limits. Efficient Agents [392] compares inference schedules through task success and token cost, DualPath [452] addresses storage bandwidth in KV-cache serving, and FreeToken [453] allocates CPU/GPU execution and cache loading on constrained hardware. Together, these studies expose the cost of model calls and moving context into computation. Background checks and maintenance draw on that capacity alongside user-facing tasks, so increasing their frequency can raise cost and response delay.

Accurate execution also depends on whether reused context is still suitable. EAL-Bench [403] tests remembered permission information, and PAST-Bench [25] reports task-stream cases where removing a persistence component improves performance. These comparisons expose the cost of inappropriate reuse. A previous success or preference does not make a procedure suitable for a changed interface, task, or grant. An external

task record may recover a missed completion event, while a user correction can expose an obsolete assumption. The repair therefore depends on whether evidence is missing, retained state is wrong, or valid state is being used outside its current scope.

6.1.4 Physical Assistance and Collaboration

Physical assistants connect decisions to world changes whose effects may be uncertain or difficult to reverse. $\pi_{0.5}$ [2] demonstrates extended household manipulation, and $\pi_{0.6}^*$ [3] uses outcome rewards and human corrections for subsequent policy improvement. MICoBot [4] negotiates and reallocates household steps with a human partner, while Learning to Assist [5] studies assistance and waiting during collaborative assembly. These choices determine the assistant’s commitments and coordination with its partner.

Retained experience and new observations support different decisions during that work. PhysMem [454] supplies retained physical experience, Being-0 [455] combines skills with active camera control, and ODYSSEY [456] uses semantic mapping for mobile manipulation. Camera control and mapping can supply evidence for the next movement and for reviewing a completed one. ABot-AgentOS [290] retains selected graph assets with provenance and reuses failure-derived assets on later evaluation splits, separating experience reuse from repairing and rescoreing the same task.

Bottlenecks and trade-offs. AssistantX [6] makes the verification problem concrete: messages, robot location, and lock state are compared with expected results before memory and plans are revised. Its reflection ablation reports improved task success alongside possible redundant behavior, and an office deployment lasting one and a half months collected 289 feedback responses. Review can help recover uncertain execution, but additional observation and replanning also consume time and resources. A reliable receipt can confirm the operation it reports. Active inspection becomes useful when physical progress or side effects remain uncertain. The user’s response and later task progress address whether the assistance was useful. The choice depends on which consequence must be checked and the cost of delaying or repeating the action.

6.1.5 Representative Systems

Table 18 brings together the four application families. One shared bottleneck is competition among continuing commitments for observation, maintenance, and intervention capacity. A familiar routine may favor reduced sampling while an unresolved action needs fresh confirmation. Giving that action priority can delay background checks on other commitments; preserving every observation instead raises storage and correction costs. Sustainable-serving studies [457] motivate budgeting for the full service workload.

Recovery can depend on the same resource or permission whose failure caused the original loss. An external task log may replace a missed completion observation, but access to both can disappear under the same permission change or outage. Whether a repair survives such shared failures is a system-level question (Section 8.2). Work on externalization [458] motivates tracing such dependencies across components. Closure levels specify what must remain available; HALO examines the evidence for dependencies and outcomes.

Table 18: **Systems across application regimes.** Groups follow the four application subsections. Each row identifies a core mechanism and a potential bottleneck. Here b is the interaction domain, n the participant scope, and h the selected case’s outcome horizon.

System	$b \times n \times h$	Key mechanism	Bottleneck
Situated Daily Assistance			
Vinci [8]	 $\times I \times s$	Streaming visual responses	Response latency / coverage
ProAssist [430]	 $\times I \times s$	Context-triggered procedural dialogue	Event interpretation
ProMemAssist [9]	 $\times I \times s$	Sensed task state guides help timing	Task-state estimation
Vinci2 [276]	 $\times I \times s$	Video memory guides when to intervene	Compression / selection
VisionClaw [10]	 $\times I/G \times s$	Visual capture, recall, and agent actions	Capture budget / privacy
Memento [220]	 $\times I \times s$	AR recall with referent/removal controls	Recall relevance / timing
Sensible Agent [431]	 $\times I \times s$	Context selects content and modality	Interruptibility uncertainty
SpeechLess [218]	 $\times I \times s$	Spatial memory guides intent inference	Stale / ambiguous state
ProAgent [179]	 $\times I \times s$	Persona-guided sensing; feedback revision	Prior error / sensing cost
Pro ² Assist [432]	 $\times I \times s$	Motion-aware sampling and step checks	Step uncertainty / repetition
LightMem-Ego [433]	 $\times I \times s$	Asynchronous consolidation for recall	Consolidation / retrieval lag
Alpha-Service [244]	 $\times I \times -$	Event triggers, memory, and tool use	Triggers / authority
Conversation and Meeting Assistance			
SocialMind [239]	 $\times G \times s$	Persona-conditioned AR suggestions	Audience / context
LlamaPIE [390]	 $\times G \times s$	Audio-guided in-ear suggestions	Turn timing / consent
AURA [406]	 $\times I \times s$	Streaming audiovisual response control	Input–output coordination
JoyAI-VL-Interaction [167]	 $\times I \times s$	Response gating; asynchronous tasks	Coordination / stale results
Thinking Machines [178]	 $\times I \times -$	Live interaction with background tools	Context limits
Digital Workspace and Personal Services			
PASK [109]	 $\times I \times s$	Intent cues and post-session memory	Intent freshness
PITCH [245]	 $\times I \times c$	Daily plans guide scheduled follow-ups	Plan relevance / receptivity
User as Code [277]	 $\times I \times s$	Typed state and executable alert rules	Schema / state correctness
ChatGPT Pulse [442]	 $\times I \times -$	Context-driven background research	Relevance / attention cost
Claude Cowork [443]	 $\times I \times -$	File access and multi-step execution	Permissions / recovery
Muse [15, 444, 68]	 $\times I \times -$	Memory, scheduling, permission gates	Authority / freshness
Windows Recall [311]	 $\times I \times -$	Protected screen-history retrieval	Admission / access control
KnowAct-GUIClaw [448]	 $\times I \times s$	Skill-guided GUI use and procedure repair	Stale skills / task contracts
MOSS [459]	 $\times I \times c$	Source repair with replay and rollback	Repair transfer
Ouroboros [460]	 $\times Pub \times -$	Reviewed memory/core self-modification	Revision / authority drift
HealthClaw [358]	 $\times I \times l$	Health-history updates guide later QA	State conflicts / privacy
Physical Assistance and Collaboration			
$\pi_{0.5}$ [2]	 $\times I \times s$	Vision-language household manipulation	Observability / execution
$\pi_{0.6}^*$ [3]	 $\times I \times c$	Feedback-driven policy retraining	Feedback coverage / transfer
MICoBot [4]	 $\times G \times s$	Dialogue-based negotiation / reallocation	Partner coordination
Learning to Assist [5]	 $\times G \times s$	Human intent gates help versus waiting	Intent uncertainty / safety
PhysMem [454]	 $\times I \times s$	Experience-guided physical manipulation	Experience transfer
AssistantX [6]	 $\times G \times s$	Cross-domain plans; outcome review	Availability / verification
Being-0 [455]	 $\times I \times s$	Skill handoff and action-guided camera	View / skill compatibility
ODYSSEY [456]	 $\times I \times s$	RGB–LiDAR maps; hierarchical plans	Map uncertainty / coverage

b :  physical,  wearable,  digital; n : I individual, G bounded group, Pub public; h : s within-session, c cross-session, l longitudinal, – unmeasured. Multiple icons or a slash retain multiple settings. Here s pools immediate and within-session endpoints. Horizons describe the selected service or post-update result, not input-history length or total deployment duration; a multi-day study can therefore have s for its selected endpoint.

Pulse, Cowork, Muse, and Recall summarize product documentation; Thinking Machines is a research preview; Alpha-Service is a demonstration; HealthClaw uses synthetic histories. Reported channel mechanisms and evidence are reviewed in Table 21.

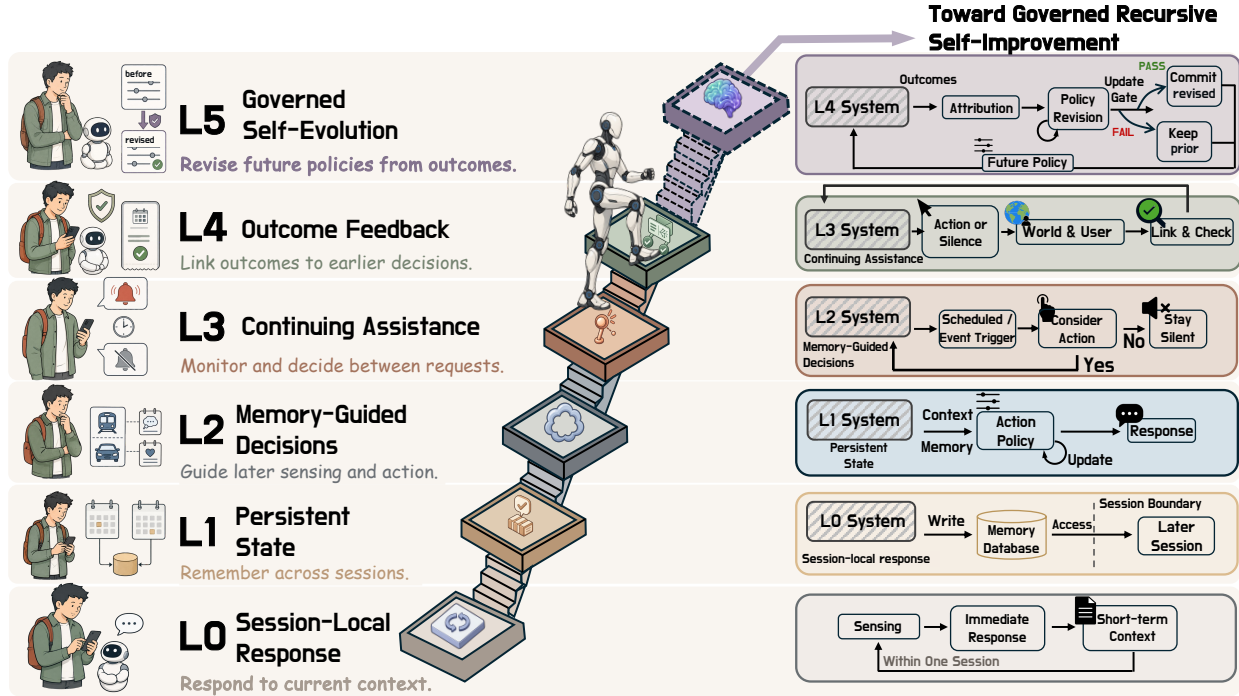


Figure 9: **Five levels of loop closure.** L0 is the session-local reference baseline; L1–L5 retain the preceding requirements and add the capability shown. L3 supplies the continuing-loop baseline; L4 connects outcome feedback; L5 supports a self-evolving loop. L1 requires retention across interactions; L2 requires retained state to inform Sensing or Action. The figure’s “across sessions” and “sensing and action” labels are illustrative.

6.2 Levels of Loop Closure

Figure 9 organizes loop closure into five cumulative levels, with session-local response as the reference baseline. Channel status is interpreted alongside the service requirements: operation between requests and validated, durable policy revision determine whether the reported mechanisms support L3 and L5, respectively.

L1 requires durable personal state across interactions. L2 requires that this state reach and inform a later Sensing or Action decision. L3 additionally provides scheduled or event-triggered monitoring and decision opportunities between requests, including waiting and silence. L4 obtains external outcome evidence, links it to relevant earlier decisions, and uses it to check or correct assistance. L5 uses attributed outcomes to propose, validate, and durably commit policy changes. Authorized tool execution is one way to provide assistance at these levels; a reminder service can close the same feedback path through user responses.

For a declared configuration, assign the highest cumulative level supported by the evidence and state what prevents a stronger judgment. Capabilities beyond that level remain separate: an invocation-only system may revise prompts while lacking L3’s continuing operation. Table 19 applies this rule to three configurations from one study. Their shared decision schedule makes missing retained personal state visible as a separate limitation. The trained configuration also leaves L5’s validation requirement unresolved. HALO then examines the mechanisms and comparisons supporting these judgments (Section 7).

Other published configurations illustrate different parts of this progression. PILOT [461] separates live steering of an active worker from persistent skill and memory updates: the supervisor can redirect the current run, while subsequently spawned workers load the revised harness. Its repeated-task evaluation examines cross-run reuse. Ouroboros [460] reports reviewed self-modification, including a duplicate-send guard used in subsequent interactions. HealthClaw [358] combines state and procedure updates. These studies make persistence, policy revision, and later use observable in different configurations.

Table 19: **Worked closure judgments in a simulated service.** ProPerSim [314] evaluates 32 personas over 14 simulated days, with recommendation or silence decisions every 2.5 simulated minutes. Levels below are our source-based judgments of configurations in that setting.

Configuration	Supported level	Cumulative support	Boundary of the judgment
No Memory	Below L1	Scheduled decisions use only the current user action.	No retained personal history: operational continuity alone does not meet the always-on definition.
AR Memory	At least L3	Retained action/recommendation history informs later scheduled decisions (L1–L3).	No explicit rating feedback or training; later observations alone do not establish outcome-linked correction (L4).
ProPerAssistant	At least L4	History and scheduling plus recommendation-linked ratings used for daily preference training (L1–L4).	Policy updates are reported, but a separate candidate-validation gate before durable commitment (L5) is unspecified.

Source: ProPerSim [314], Sections 3, 5, 6.1–6.2, and Table 1. Feedback is immediate scoring by a simulated evaluator. These judgments establish neither real-user benefit nor a deployment-wide level; unspecified requirements remain unknown.

Functional loop completeness concerns whether the available mechanisms support the dependencies needed for the declared service. *Reporting completeness* concerns an explicit status for all five channels. A complete report can therefore describe a lower-level or incomplete service. The levels organize closure requirements; service quality requires outcome comparisons under matched conditions.

Diagnose the Dependency Before Choosing the Remedy. The same visible error can come from missing observations, obsolete personal state, inappropriate use, or misattributed feedback. The application regime identifies the relevant costs and permissions; tracing the service across channels identifies where a repair belongs. L1–L5 then organize progress from retained state through continuing assistance to governed revision, with evidence attached to each added requirement.

7 HALO: Harness for Always-on Loop Oversight

Section 6 identifies application bottlenecks and closure requirements. HALO supports two complementary modes under a common oversight protocol: *read mode* for source-based review and *run mode* for scoped execution. Both assess channel mechanisms, comparisons, and service outcomes against these requirements. Figure 1 shows the two workflows; Section 7.3 presents the read-mode review of existing systems.

7.1 Operating Modes and Evidence

Read mode and run mode. Read mode inspects papers and available code, traces, and other artifacts without executing the target. It produces source-linked channel judgments and assesses comparisons reported by the original study. Run mode extends this review with isolated tests of an executable configuration. A suitable interface or scoped adapter connects the target to specified initial states, scenarios, and supported interventions. Execution produces manipulation checks, traces, and measured endpoints. Both modes use the same evidence criteria, while source-reported and newly executed results retain separate provenance.

In read mode, the review unit is the configuration evaluated in the source; in run mode, it is the executed configuration. Reset rules, disabled capabilities, and interaction boundaries determine which mechanisms are available during the comparison under review. An evaluation that resets memory is analyzed in that configuration even if the original platform supports persistent state. HALO distinguishes four forms of supporting evidence, several of which may be supplied by one study:

- **Mechanism description:** the source identifies the information source, receiving decision, coupling mechanism, and evaluation scope.
- **Traced use:** run records show the delivered information, policy decision, active conditions, and downstream use.

- **Controlled effect:** a controlled comparison changes a specified input, policy, or component and measures a downstream decision or state.
- **Controlled service outcome:** a controlled comparison measures utility, user cost, safety, recovery, or another service endpoint over a stated horizon.

These forms describe support for a claim, circles record mechanism status, and L-levels describe closure requirements (Section 6.2). The four forms of evidence are not cumulative: a controlled comparison can confirm the manipulation through its design or records without publishing full run traces. Each comparison retains its manipulated unit, setting, horizon, and reported uncertainty, distinguishing a channel effect from a whole-pipeline effect. A short study, simulated setting, or missing ablation concerns evaluation scope and does not make an implemented mechanism partial.

Source records and traces. Each record identifies the paper version, configuration, service scope, source location, channel judgment, and comparison supporting it. It distinguishes the authors’ description from the reviewer’s functional interpretation. When sources provide traces, the review follows observations and retained state through delivered signals, receiving decisions, and subsequent updates. Timestamps, policy versions, and active conditions connect these steps. Direct user feedback, tool-reported outcomes, sensed world changes, and internal assessments retain distinct provenance. Capture and suppression records also explain evidence availability and incurred costs, connecting persistent-state lifecycle reporting [57] to channel use under shared conditions.

Mechanism status and coding. Four symbols summarize mechanism status: ● indicates a reported implementation, ◐ an implemented part with an explicitly unfinished requirement, ○ an explicitly absent mechanism, and ◑ insufficient reporting to determine the dependency. A partial judgment names both the completed and unfinished parts, while an absence judgment cites the source’s exclusion. Retention is assessed across the source’s user interactions, including later exchanges within a day. When task resets leave personal-state lifetime unclear, the record preserves that ambiguity.

Coding follows the information source to its immediate receiving decision. A retained identity can directly inform interpretation of a new clip. Retained preferences can instead determine which question is asked, which then determines the next observation. The latter receives a star, as in ●*, with a source record tracing the linked decisions and endpoint. Finding two channels separately or a fused controller is insufficient to confirm this connection. A service can also benefit from other evidence: a delivery receipt can settle whether a message arrived even when no action-conditioned sensing check is reported. The review retains the source-supported $A \rightarrow S$ status and identifies the service result supported by the receipt. Stars require source support.

7.2 Channel Tests and Outcome Assessment

In read mode, HALO identifies what a published comparison changes and what it holds fixed. In run mode, the same comparison design guides isolated tests where the required interfaces are available. A channel-targeted comparison changes one payload or coupling policy while matching the relevant local capability, initial state, and shared conditions. The trace locates the manipulated signal and receiving decision, and the comparison measures the downstream endpoint. Table 20 guides the interpretation of published experiments and the design of additional run-mode tests.

Removing a memory service or controller measures several dependencies jointly. Narrower comparisons distinguish an upstream state change from replacement of the signal delivered after retrieval; their causal interpretation depends on what each preserves and changes [462]. Stale and misleading inputs also differ from missing inputs: memory pollution [463] and authority laundering [403] can propagate through a functioning persistence route. Independent verifier checks address another source of error, since self-authored tests may reward behavior that fails external evaluation [464].

Settings and controls. Matched scenarios preserve the initial personal state, available observations, request, permissions, and resource constraints, with controlled randomness where feasible. Offline tests and replay compare fixed inputs or recorded temporal context, while shadow mode records proposed interventions

Table 20: **Coupling-channel tests and diagnostic evidence.** Match the service, initial conditions, and relevant capabilities. The endpoint follows the manipulated interface; the final columns distinguish possible faults and the evidence needed to resolve them.

Channel	Paired change / control	Receiving decision / endpoint	Failure to distinguish	Evidence to resolve it
S → M	Alter the admitted write y_t^{SM} ; preserve memory update and later read policies.	π_M : committed state, then later availability and use.	Evidence omitted before admission; rejected write; failed persistence.	Proposal and admission records, state diff, later read; independent coverage reference for missed evidence. Write/compression example: Imprint [171].
M → A	Replace delivered y_t^{MA} ; fix store, retrieval, sensed state, authority, and executor.	π_A : timing, content, preparation, execution, or silence.	Wrong retained state; wrong delivery; inappropriate use.	Retrieved and delivered records, consumer trace, action endpoint. State lesions: User as Code [277]; authority swaps: EAL-Bench [403].
A → S	Change y^{AS} for the same intended/executed action; fix sensing capability.	π_S : information acquired before acting or checks of effects afterward.	Unspecified observation need; failed acquisition; reliable receipt already sufficient.	Target, acquired evidence, latency, recovery and cost. Camera control: Being-0 [455]; review: AssistantX [6].
M → S	Replace y_t^{MS} ; fix the available stream and sensing budget.	π_S : sampling, attention, or interpretation; later assistance.	Stale prior; missed counterevidence; ambiguous current observation.	Prior and sensing trace; coverage reference or interpretation labels; matched cost. Guidance examples: ProAgent [179], SpeechLess [218].
A → M	Freeze, delay, or misassign y_t^{AM} ; fix observed outcomes and prior trace.	π_M : personal-state revision; governed updater: policy revision; later decisions.	Wrong outcome link; unjustified credit; failed update; harmful revision.	Outcome provenance and competing causes, update diff, later controlled outcome. Skill repair: KnowAct-GUIClaw [448].

The cited studies motivate the controls; each comparison still needs an explicit intervention location. A traced payload may have no detectable marginal effect. For A → M, update influence, justified attribution, and later benefit are evaluated separately.

without executing them. Simulation supports controlled reruns, and field deployment can observe actual user adaptation, trust, and fatigue. The setting determines which outcomes are visible; the evidence judgment still depends on the comparison. For example, supplying or withholding a reliable completion receipt tests the conditions under which active checks remain necessary.

Controls first check that the intended manipulation occurred, then interpret the detected or undetected effect. No detected change can reflect an inactive dependency, redundant evidence, compensation, or limited test sensitivity. Matched repetitions and controls expected to have no effect help distinguish these cases from test variation. Uncertainty is aggregated over independent scenarios or users, avoiding treating correlated events within one trace as independent samples. Joint comparisons further examine compensation and error propagation under the same budget.

Reviewing attribution under delayed feedback. Consider an analytical example: a reminder followed later by a task-completion receipt. The task may have been completed because the reminder helped, because the user acted independently, or because an external event resolved it. The same receipt can be compatible with all three explanations. A comparison that freezes the resulting memory update can reveal an effect on later reminders while leaving the original cause unresolved. HALO therefore distinguishes the outcome association, its role in the update, the evidence supporting attribution, and the later service outcome.

A reliable receipt supports updating task status, and an explicit user correction can support revising the named preference. Crediting the reminder with causing completion requires evidence that distinguishes competing causes. Attribution accuracy needs a reference cause or controlled generating process; several compatible causes leave attribution uncertain. The example thus suggests a further test: hold the visible outcome fixed while varying a known generating cause, or collect feedback that distinguishes the causes. This is a test design derived from the protocol.

Service outcomes over the horizon. Write and intervention counts describe system activity; service value depends on its effects over the relevant horizon. Audits of personal-agent protocols [465] motivate coverage of changing preferences and context, while timing studies [466, 13] show how selected decisions can be compared. A fixed-policy service needs evidence of sustained quality. A self-evolving service also needs attributed revisions, independent acceptance checks, and later benefit over a suitable unrevised policy.

Deployment extends the outcome set to fatigue, trust, correction and deletion effort, bystander effects, and recovery after mistakes. HealthClaw’s synthetic trajectories [358] and Ouroboros’s deployment record [460] illustrate different evidence sources. HealthClaw’s matched comparisons also change backbones, while Ouroboros’s 161-day deployment record and frozen-configuration benchmark campaigns provide separate evidence. Simulated trajectory length, elapsed deployment time, and the delay to a consequence describe different temporal properties. Each service conclusion is therefore linked to the comparison and temporal scope supporting it.

7.3 Read-Mode Review of Existing Systems

Table 21 applies HALO in read mode to 31 configurations from 30 works; empirical results belong to the original studies. The inventory re-reviews the purposive system matrix from the first inventory release (v1 ledger), using the current coding guide and retaining named service implementations, bounded task configurations, and product descriptions at their reported scope. Five module-only contributions remain in the module discussions rather than becoming system rows, and one work is split by backend. This is a narrative selection, without a search-complete candidate pool or field-prevalence estimate. Its source/evaluation groups complement the application families in Table 18.

Three coding-agent tasks reviewed disjoint paper groups under a shared guide, followed by coordinating review of ambiguous cases. The release at `analysis/evidence/release-v2.json` links the guide, source ledger, and generated views, retaining each cell’s rationale, realization, source location, and reconciliation decision. The selection record documents these exclusions.

Table 21: **Read-mode review of personal-assistance configurations.** ● reported implementation; ○ explicitly partial; ◐ explicitly absent; ◑ unreported or insufficiently specified. An asterisk marks realization through a documented combination of existing channels.

System / configuration	S→M	M→A	A→S	M→S	A→M	Mechanism / comparison scope
Contextual assistance, personal recall, and continuing services						
Vinci2 [276]	●	●	◐	◐	◐	Memory/retrieval ablations; four time horizons
VisionClaw [10]	●	●	●	●*	◐	Capture/recall; delegated-service composition
ContextAgent [243]	◐	◐	●	◐	◐	GPS-to-weather planning; retention unspecified
ProAgent [179]	●	●	●	●*	●	Persona-guided sensing; feedback/sampling tests
AiGet [240]	●	●	◐	●	◐	Profile/focus selection; feedback refinement absent
Alpha-Service [244]	●	●	●	◐	◐	Cross-session context; orchestration demos
Memento [220]	●	●	●	●	●	Contextual recall across days; targeted removal
SocialMind [239]	●	●	◐	●	◐	Persona/nonverbal comparisons; separate live study
SpeechLess [218]	●	●	●	●	●	Spatial intent/confirmation; effort/usability studies
M3-Agent [130]	●	●	◐	●	◐	Entity memory; ablations in bounded video QA
LightMem-Ego [433]	●	●	◐	◐	◐	Async consolidation; recall QA and latency
PASK [109]	●	●	◐	◐	◐	Intent/memory; latency tests, small user studies
User as Code [277]	●	●	◐	◐	◐	Typed state; extraction-to-alert comparisons
PITCH [245]	●	●	●	●*	◐	14-day field study; same-day recall only
Ouroboros [460]	●	●	●	◐	●	Reviewed repairs; live traces, separate frozen tests
AssistantX [6]	●	●	●	●*	●	Reflection; sim. ablations + 1.5-month office use
Bounded activities and simulated task streams						
Sensible Agent [431]	◐	◐	●	◐	◐	Adaptive prompts; burden/time vs. voice queries
Egocentric Co-Pilot [139]	●	●	●	◐	◐	History/tool tests; trials and blinded log ratings
VisualClaw/Claude Code [187]	●	●	●	●*	●	Native inspection; Arena correctness/cost
VisualClaw/Codex [187]	●	●	◐	◐	●	Snapshot/write bridge; Arena correctness/cost
Pro ² Assist [432]	◐	◐	◐	◐	◐	Step-aware guidance; ablations + 20-user study
Plan, Watch, Recover [230]	●	●	●	●	◐	Plan-guided checks; planner/duplex + plan tests
IPI-Agent [241]	●	●	●	●	◐	Editable requests; intent routing and gating
Proactive Agent / Gym [12]	●	●	●	●*	◐	Simulated assistance; separate proactivity tests
PAHF [221]	●	●	●	●*	●	Pre/post/both feedback; simulated preference drift
AwareLLM [14]	●	●	◐	●	◐	Preference/dialogue memory; calibration in Sensing
ProAct [238]	●	●	●	●*	◐	Idle-time acquisition; preparation/memory tests
Long-term TOA [338]	●	●	●	●	●	Task triggers; ChronosBench in simulated time
ProPerAssistant [314]	●	●	◐	◐	●	History/ratings/DPO; simulated recommendations
Background service products: documentation						
ChatGPT Pulse [442]	●	●	◐	●	●	Daily research; retained personalization feedback
Muse [15, 444, 68]	●	●	●	●*	◐	Persistent tasks; permission-gated browser sensing

VisualClaw rows denote Arena backends; Gym denotes Environment Gym. Starred cells here denote memory guidance of sensing through action decisions; the ledger records the linked steps. Status describes implementation; comparison scope and evidence supporting each claim are recorded separately. Source locations, retention boundaries, and update branches are recorded in Section 7.1 and the v2 ledger.

What the reported mechanisms cover. Within these 31 configurations, $S \rightarrow M$ and $M \rightarrow A$ each have 28 reported implementations, $A \rightarrow S$ and $M \rightarrow S$ each have 19, and $A \rightarrow M$ has 11. Of the 19 $M \rightarrow S$ implementations, nine use a documented $M \rightarrow A \rightarrow S$ composition, six directly condition the sensing decision, and four fuse memory and observation in a shared computation with an identifiable receiving decision. These are alternative realizations of a dependency, not independent experiments. $A \rightarrow M$ has 18 insufficiently reported cases and two explicit absences. The inventory thus documents retention and reuse more fully than feedback into persistent updates, locating a reporting gap without converting unknowns into absences. ProAgent [179] compares feedback-derived personas and sensing choices, while Memento [220] supports targeted record removal: both incorporate feedback, but only the former comparison examines its effect on later assistance-triggering decisions.

Reviewing a published feedback mechanism. For the ProPerAssistant [314] configuration in Table 19, HALO separates the dependencies behind the L4 judgment. It retains user-action and recommendation history ($S \rightarrow M$) and retrieves earlier interactions when choosing a recommendation or silence ($M \rightarrow A$). Recommendation-linked ratings form preference pairs for daily DPO updates, supporting the policy branch of $A \rightarrow M$. The source does not specify $A \rightarrow S$ or $M \rightarrow S$ (○).

The comparisons distinguish no history, interaction history, history with ratings, and preference-trained assistance. Adding ratings to context and using them for training change the decision input and update mechanism, respectively. These controlled service outcomes cover recommendation quality, timing, and frequency across simulated days, supporting adaptation to the simulated evaluator. Delayed effects on real users require corresponding user and temporal settings.

Which claims the comparisons support. AiGet [240] compares its full pipeline with a variant that keeps profiles but omits the selection rules and a variant with neither profiles nor selection rules. Adding profiles alone improves perceived personalization without significantly improving overall ratings; combining profiles with focus and familiarity rules improves knowledge ratings. Because these rules depend on profiles, the study has no rules-only condition. These comparisons support the combined benefit of profiles and rules, while the contribution of one channel requires a more targeted comparison.

Runtime configuration determines which dependencies are available. VisualClaw [187] reports two Arena backends: Claude Code reads and checks the workspace through its tool loop, whereas the paper’s configured Codex bridge receives a snapshot before the call and applies emitted writes afterward. The native tool loop can acquire evidence during execution, while the snapshot bridge excludes that route; both are scored externally. This supports different $A \rightarrow S$ judgments for the two configurations. The reported no-evolution and sampling comparisons evaluate correctness and cost within each backend; differences between backends do not isolate a single channel’s effect.

These comparisons motivate two priorities in Section 8. First, a reported $M \rightarrow S$ mechanism needs tests of whether corrective observations remain available when retained priors become stale, especially under the shared constraints identified in Section 6.1.5. ProAgent [179] provides sensing comparisons, and PAHF [221] supplies simulated preference changes; neither examines whether corrective evidence survives when observation capacity and access fail together. Second, the sparse $A \rightarrow M$ reporting calls for outcome-linked records and explicit acceptance evidence before expanding policy search (Section 8.3). Better search cannot recover an unobserved consequence or establish why a revision was warranted. The priority is to make continuing assistance observable and correctable, then test which revisions improve its declared outcomes.

7.4 Reporting and Limitations

Reproducible materials and system card. The review record preserves sources and judgment rationales. Available experimental artifacts further connect model, prompt, policy, and environment versions to initial states, traces, outcomes, interventions, and metrics. Aggregation units and failed or unchanged outcomes preserve the result’s scope; redacted examples and source hashes support checking while protecting personal data. Operating regimes determine additional detail: sensing configuration, retention, and consent for wearables; permissions, confirmation, and rollback for digital action; admission, retrieval, correction, and deletion for memory; and turn boundaries, audience, interruption, and silence for conversation. Each detail connects to the declared service and outcome horizon.

Table 22 summarizes the review output for both module contributions and complete-system claims. Mechanism status identifies the dependencies covered by a report; comparisons and outcomes determine the service judgments they support.

Table 22: **Minimal system card.** Report the service, mechanisms, and evidence needed to assess the system’s commitments.

Field	Required content
Service and regime	Purpose; (b, n, h) ; availability and state durability; intended outcomes; any adaptation claim.
Spaces and local policies	Available observations, retained objects, and effectors; sampling, memory lifecycle, and act/silence policies.
Five channel statuses	Reported, partial, explicitly absent, or unreported; mechanism and source location; the mechanism behind any asterisk and service-specific exclusions.
Shared conditions	Privacy, resources, uncertainty, and user authority; participant consent, interruption, reversibility, and suppression rules.
Configuration and traces	HALO mode; paper/artifact and model/harness versions; available state and decision records; outcomes, attribution, and revision links.
Comparisons and outcomes	Evidence, comparison unit, and endpoint per claim; single/joint interventions; compensation; utility, cost, safety, recovery, setting, horizon, and uncertainty.
Revision and release	If evolving: candidate, acceptance evidence, committed version, later effect, rollback; otherwise fixed-policy scope. Link available artifacts, verifier, and failed/unchanged cases.

Limitations. HALO reviews depend on the detail and accuracy of the sources. Functional mappings may need reconciliation when implementations fuse decisions, and unreported mechanisms remain unresolved. Agreement between reviewers still needs duplicate independent coding. Existing comparisons support conclusions within their configurations and horizons, and the shared protocol makes those judgments checkable. The present synthesis covers read-mode findings; cross-system validation of run mode requires executable configurations and appropriate controls. Further independent review and longitudinal evaluation can test coding consistency and the reach of the identified service claims.

8 Trends and Open Problems

The research agenda follows the dependencies exposed by Sections 6 and 7: preserve opportunities to observe change, keep retained state correctable, and establish the warrant and later benefit of policy revision. This is a research order of dependence, not a ranking inferred from channel counts. Fixed-policy services can qualify as always-on; self-evolution needs evidence that later benefits justify its costs.

8.1 Local Policies for Continuing Assistance

A continuing assistant needs to detect when its usual observation policy has become inadequate. The open question is when to revisit familiar situations and how to represent uncertainty about intervals that were not observed. Periodic exploration and uncertainty-triggered checks may recover correction opportunities at additional acquisition and privacy cost. Comparisons need to track missed events and their effects on later decisions under matched acquisition budgets, including details lost through segmentation or compression.

For personal memory, the open problem is how to propagate correction and deletion through partial or compressed records without retaining every source indefinitely. Preserving episodes supports reinterpretation; selective writes reduce retention but may discard what a later contradiction requires. Preference changes and deletion requests further require distinguishing residual copies of invalidated state from information independently acquired elsewhere.

A welcome intervention today can contribute to fatigue after repeated use. Proactive-assistance and abstention studies offer ways to choose among silence, communication, preparation, and authorized execution [42, 43], but their relative value changes with receptivity and accumulated interruption. The open problem is how initiative should adapt to delayed benefit, missed opportunities, and cleanup effort. Reward-design work such as ProRL [467] motivates examining whether objectives that encourage activity also improve the user’s outcome. Frequent acceptance alone leaves that longer-term tradeoff unresolved.

8.2 Coordination Across Sensing, Memory, and Action

The contention identified in Section 6.1.5 raises a local coordination problem: how should π_S reconcile memory guidance through $M \rightarrow S$ with an action’s observation request through $A \rightarrow S$? Candidate policies could prioritize fresh evidence for unresolved effects or combine requests with compatible views, modalities, and freshness requirements. Under a fixed acquisition budget, comparisons should vary agreement and conflict between the two inputs, measuring evidence freshness, task completion, and missed correction opportunities. This tests their interaction beyond the effect of adding either input alone.

When several tasks overlap, concentrating observation on one may reduce opportunities to detect new events or follow other pending commitments. Research is needed on how task-specific checks coexist with background observation, and how their demands are released or redirected when a task ends, changes, or is cancelled. Evaluations with overlapping tasks and newly arriving events could compare current-task success, missed opportunities, evidence freshness, and total acquisition cost. This would test whether focused observation remains useful across the continuing service.

A task log or broader monitoring may recover evidence omitted by selective sensing, but obtaining that evidence can depend on the same constrained resource or permission. The open question is when compensation remains available if several dependencies are affected together. Comparing the original loss and its recovery across resource and access conditions would reveal which remedies remain effective and whether their benefit justifies the added cost.

8.3 Self-Evolution and Policy Revision

Unresolved errors can recur across an always-on assistant’s interactions. Learning from a user’s dismissal or correction requires linking that feedback to earlier decisions and determining whether to correct personal state or revise a decision rule. From L4 to L5, the open question is which attributed outcomes justify policy revision and how its benefit can be established in later interactions. Policy revision can weaken existing capabilities: Don’t Blind Your VLA [468] finds that action fine-tuning can degrade visual-language grounding, motivating checks of perception alongside task performance.

Search supplies candidates; acceptance decides which to retain. PACE [80] and certificate-gated evolution [81] examine acceptance under repeatedly inspected noisy scores. Self-authored verification [464] and memory misevolution studies [86, 87] identify failures of insufficiently checked scores or reuse. Memento-Skills [469] revises skill files from failure traces, Zetta [470] repairs critics or recovery code, and StateM [471] versions runbook practices. Table 23 compares, per configuration, the updated object, the acceptance or activation rule, and the evidence of later use; Table 24 maps complementary research by role.

Recursive revision changes the update mechanism. Policy revision at L5 can use a fixed updater $\mathcal{U}_{\psi_t}$ and verifier V_{ϕ_t} . Recursive revision additionally changes one of these mechanisms, retains the change, and uses it to govern a later improvement round [505]. It therefore requires at least:

$$\text{recursive at } t \implies (\psi_{t+1} \neq \psi_t) \vee (\phi_{t+1} \neq \phi_t). \quad (5)$$

The relevant evidence follows the revised mechanism into its next use. Recursive Harness Improvement [493] uses revision history, while Metaⁿ [506] keeps the meta-operation fixed. Hyperagents [507] reuses evolved meta agents to generate subsequent variants. Its transfer tests hold the meta agent fixed to examine the improvements it produces in a new task domain; parent selection remains fixed in the main experiments. The Red Queen Gödel Machine [508] instead evolves the evaluator: it freezes the evaluator within each search epoch, selects replacements against a fixed ground-truth anchor, and discards scores that depend on the replaced evaluator. The successor evaluator then governs later search. Offline meta-learning instead learns memory designs or updaters before deployment [509, 510]. These cases distinguish repeated policy revision from inherited changes to the process that produces later revisions [45, 511]. User authority and deployment conditions continue to constrain both.

Table 23: **From feedback to an active update.** Selected configurations separate the updated object, the acceptance or activation rule, and the evidence of later use.

Configuration	Updated object	Acceptance / activation	Evidence and boundary
Memento [220]	Personal record	User selects an unwanted memory for permanent dismissal from recurring recall.	Removal control and diary study; record correction does not establish policy revision.
ProPerAssistant [314]	Model weights (LoRA)	Recommendation ratings form preference pairs for daily DPO; a separate pre-commit validation gate is unspecified.	Simulated recommendation outcomes; closure level and its limits: Table 19.
EmbodiSkill [322]	Skill body and appendix	Reflection distinguishes skill defects from execution lapses. After consolidation, revised skills activate at the next stage without an outcome-based acceptance test.	Evolved skills are fixed for held-out evaluation. Test performance does not itself describe an online acceptance gate.
PACE [80]	Prompt candidates	Paired candidate/incumbent evaluation; commit when an anytime-valid test reaches its threshold.	Held-out accuracy on three reasoning benchmarks in one model family; false-commit control is per candidate under the test assumptions, not a run-level bound.
MOSS [459]	Harness source	Review and failure-batch replay precede user-authorized container replacement; a health check governs rollback.	External grader scores on the same four-task benchmark batch used for repair, with consent auto-acknowledged; transfer to later workloads is untested.
Ouroboros [460]	Core code and prompts	Preflight, staged-diff fingerprint, blocking review, and fingerprint recheck before commit; scope review depends on context mode.	Reviewed changes become the later runtime. Live deployment and frozen benchmark campaigns use separate lineages.

Source locations: Memento Sections 2.2.5–2.2.6 and 4.2; ProPerSim Sections 5–6; EmbodiSkill Sections 3.2 and 4.1; PACE Sections 3–6; MOSS Sections 3–4; Ouroboros Sections 3–5. Mechanism descriptions and measured improvement remain separate claims.

8.4 Longitudinal Evaluation and Real Deployment

Longitudinal evaluation must preserve the connections among unresolved commitments, recovery after pauses, preference drift, and delayed outcomes. Such comparisons face a changing feedback distribution even when the evaluation rule remains stable. Performative prediction studies how deployment changes later data [512], while recommender simulations examine feedback and algorithmic confounding [513]. Personal assistance likewise changes habits, receptivity, and subsequent feedback. Freezing the software judge, rubric, and each arm’s policy version stabilizes the comparison procedure, but the user can still change in response to earlier interventions. DR Tulu [514], RubricBench [515], and studies of changing verification [516, 517] offer relevant evaluation designs. Fixed-input replay measures sensitivity to recorded evidence; closed-loop or deployment comparisons are needed for effects mediated by user responses.

Policy revisions need to be evaluated in later interactions with independently checked outcomes, beyond the tasks used to select them. PAST-Bench [25], PATH-Bench [351], and SEA-Eval [72] provide task-stream mechanisms for examining such effects. Comparisons between fixed policies, revision under fixed updaters, and revision with mutable updaters should match initial states, available evidence, and total budgets, including revision search and validation costs. For recursive claims, better current-task performance must be distinguished from an improved ability to produce or select subsequent revisions. Evaluation should also examine when search is paused or ended as its expected benefit falls below the cost or risk of further revision [505]. Versioning records the mechanism behind each update and supports rollback. The service endpoint remains the declared user benefit, cost, or tradeoff.

Table 24: **Complementary research on persistent updates.** Separate panels organize update substrates, digital instantiations, and evaluation or governance; these are different research roles, not one taxonomy axis.

Research line / focus	Representative works	Question for continuing personal service
A. Mechanism sources, grouped by updated substrate		
Experience and memory Episodes, graphs, utility slots, latent state, retrieval configuration	ExpWeaver [318], MemQ [302], RoMeRL [472], SAGE [473], R ² -Mem [474], FORGE [475], EXG [476], EvolveMem [325], OPD-Evolver [323], MoLEM [477], Auto-Dreamer [300], DCPM [301], LiveMem [478], and PGMem [479]	Which records remain usable across interactions, and how are source links and corrections preserved?
Skill lifecycle Text/code skills, catalogs, curation histories, structured skill graphs	SkillOS [480], SkillsVote [320], raw-experience lifecycle study [481], SkillOpt [482], ACE [84], MUSE-Autoskill [483], SkillBrew [484], SkillComposer [485], SkillHone [486], SkeMex [487], Ratchet [488], and HyperSkill [489]	Which feedback warrants editing or retiring a procedure, and when should valid guidance remain unchanged?
Harness, tools, and code Prompts, control flow, tools, source, versioned software body, optimization meta-policy	Continual Harness [490], HASP [319], SkillSmith [491], Adaptive Auto-Harness [492], Recursive Harness Self-Improvement [493], MOSS [459], SIA [494], OurArk [495], and Open-Ended Optimization [496]	Which checks cover later workloads, and how do commitment, rollback, and current authority interact?
Parameters and hybrid state Lightweight modules or model weights, sometimes with harness updates	History-to-State [497], Language Models Need Sleep [498], SEAL [499], TMEM [324], and SIA [494]	Can personal state be identified, corrected, or deleted without damaging shared capability?
B. Application instantiations		
Digital/GUI instantiations GUI memory, policy, continuation skills, mobile curricula	SE-GA [449], SCALE [450], SGCD [446], SkillHarness [447], Darwin Mobile Agent [451], and VisualClaw [187]	Which initiation, persistence, and permission mechanisms belong to the implementation rather than the testbed?
C. Evaluation and governance		
Evaluation protocols Ordered episodes, held-out task families, targeted failures, user-state interventions	EvoMemBench [353], BenchTrace [354], StreamMemBench [362], LifeSide [500], M ³ Exam [501], PAST-Bench [25], GDPevo [502], ContinualSkillBench [355], SEA-Eval [72], and temporal-intervention evaluation [465]	Do comparisons measure later use, service effects, and user response beyond recorded task sequences?
Failure and governance S → M boundary, consolidation, library, retrieval gate, retained capability	Capability erosion [363], faulty consolidation [303], state contamination [364]/poisoning [365], library drift [503], persistent sycophancy [366], stale dependencies [308], MemConflict [285], MemGate [504], and harness update/benefit separation [82]	Does invalidated guidance cease influencing later decisions, including through derived records and procedures?

The questions are our synthesis. Inclusion identifies relevant mechanisms or tests, not an established always-on configuration or a measured deficiency in every cited work.

Longitudinal data needs to link observations, decisions, and delayed outcomes under declared capture and retention boundaries [125, 296]. Recording concurrent observation demands and task schedules would allow resource sharing to be compared, while reusable intervention tools would support service studies and evidence for policy revision. Versioned sources and reusable redacted scenarios would help these comparisons accumulate across systems.

9 Conclusion

Always-on personal assistance requires operational continuity between requests and usable personal state across interactions. We analyzed this baseline through Sensing, Memory, and Action, their dependencies, and cumulative closure requirements. Across the reviewed configurations, state retention and use are more fully reported than feedback entering persistent updates. Whether memory-guided observation detects changes contradicting retained priors also remains insufficiently tested. HALO’s read-mode inventory makes these judgments inspectable; its run mode supports additional tests of executable configurations.

Progress requires keeping corrective evidence obtainable and following corrections into later decisions. Fixed-policy services can meet the always-on baseline. Self-evolving services additionally need proposals justified by attributed outcomes, validation before durable commitment, and evidence of later benefit over an unrevised policy. Both should be evaluated by user benefits and costs over the relevant horizon, within resource, privacy, and authority constraints.

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